

Energy efficiency of quantum computers

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Abstract

How much energy does a quantum computer consume? Are they more efficient than their classical counterparts? In this work, we make a step towards answering these questions. We define the energy efficiency of a quantum computer as the ratio of the number of algorithms it can perform during a given time over the energy consumed by the hardware during this time. We analyze the most representative physical platforms currently envisioned to be used as building blocks of quantum computers: superconducting qubits, silicon spin qubits, trapped ions, neutral atoms and photonic qubits. Including insights from experts in all these technologies and taking into account algorithm compilation constraints, we discuss the advantages and inconveniences of each platform from an energy standpoint. Beyond providing concrete values of the energy consumption of current quantum computers, we lay the foundation of a framework to benchmark the energy efficiency of any future quantum computing architecture.

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1 Introduction

Quantum computing is emerging as a technology of the near future, with promising applications in various fields ranging from many-body physics simulations for molecule synthesis or nuclear structure analysis, to solving optimization problems for logistics. It enables a fundamentally different computing paradigm from the so-called *classical* current computing platforms, by encoding information in the degrees of freedom of quantum particles. Several physical systems are envisioned to create these quantum bits, and it is not clear today which of them will be the key components of future quantum computers.

While quantum computing presents us with groundbreaking opportunities, it also poses the challenge of environmental sustainability. As the world struggles with the climate crisis, the construction and energy cost of future technologies has become a critical concern [1]. More and more stringent resource constraints urge the community to understand the cost of quantum computing. It can be that a particular technology, while currently being very efficient at running quantum algorithms or storing quantum information, is not scalable in a sustainable way and thus has to be discarded. It is important to study this early on, so that *path dependence* effects, where past events or decisions constrain later developments, are avoided. On the other end, the discovery of an energy advantage, where a particular task is solved using less energy than a classical counterpart, could speed up the development of a particular platform energy-wise.

In this paper, we make a first attempt at understanding the energy footprint of current quantum computing platforms. Thanks to insights from our diverse author list, we established a list of the main platforms envisioned to scale up to a large-scale and fault-tolerant quantum computer, capable of successfully executing quantum algorithms and therefore, run useful applications [2]. While several paradigms and degrees of freedom can be employed to manipulate quantum information on a particular platform, we provide an estimation of the power consumption of near-future noisy intermediate scale quantum (NISQ) computers. We discuss the possibility of extracting trends and study the scalability of each platform from an energy standpoint. Our goal here is not to rank the different platforms, but rather to understand what the critical components are and identify pros and cons of each platform.

To this purpose, we define an energy efficiency metric for quantum computer architectures and analyze their performances. For each particular platform, we listed the important hardware components as well as their energy consumption. Our analysis shows that, although executing specific algorithms can increase the classical control overhead, the total energy consumption of the quantum system does not increase significantly when transitioning from an idle (passive) state to active operation. This is largely because major hardware element, such as cryostats and laser systems, remain continuously powered, even when no computation (*i.e.* algorithm) is being performed. In our analysis, instead of focusing on a task-specific energy comparison across platforms, we evaluate their general computational capabilities and compare it to their energy consumption. This requires the derivation of different compilation methods for quantum algorithms running on each of the platforms considered.

We find that, as could be expected, most of the energy expenditure in NISQ quantum computers stems from maintaining a stable qubit environment, through cooling or temperature control. It can be noticed however that, should current architectures scale up without new fundamental discoveries and innovative engineering, the consumption related to qubit control electronics will rapidly outgrow this cost. Beyond establishing a baseline list of components for NISQ computers, our analysis studies the main constraints on the running time of quantum algorithms on each platform: qubit connectivity in the case of superconducting and spin qubits, number of interaction zones and shuttling time in the case of trapped ions and neutral atoms, and probabilistic operations in the case of photonic qubits. Our work shows that operational times of the devices, such as two-qubit gate time or the reset time between different shots of a quantum circuit, are critical parameters in improving the energy efficiency.

Using our methodology, recent techniques under investigation to improve operational times can be compared to benchmark their impact on the energy efficiency. As examples, we study active reset on superconducting platforms, qubit connectivity in spin qubit platforms, number of interaction zones in trapped ion platforms, continuous reload in neutral atom platforms and different materials for chip-based photonic computing. These results can guide policy-making and engineering decisions in the near future to improve the sustainability of quantum computers.

We note that the study conducted in this paper is only one of the building blocks required to construct

a complete view of the future energy cost of quantum technologies. One key cost that is not covered and might prove to be a dominant one, is the manufacturing cost. Materials used in the construction of quantum devices are present in finite quantities, for example rare-earth elements or Helium 3. The direction of current technological development is not concerned with the re-usability of devices and, when the state of the art advances, the previous generation of quantum computers is usually discarded. A full life-cycle assessment of quantum computers might give very different outcomes than the ones we present in this study.

Additional factors can be considered to provide a comprehensive understanding. The rebound effect for example, which refers to the phenomenon whereby improvements in the efficiency of a technology reduce the effective cost of its use, leading to increased consumption that partially offsets the expected resource or energy savings. This response can occur at multiple levels, including direct effects (increased use of the more efficient technology) and indirect effects (reallocation of saved resources to other consumption activities). Consequently, the net environmental or energy benefits of technological innovations may be smaller than anticipated without accounting for these behavioral and systemic responses.

This paper is organized as follows. Our methodology is first introduced in Section 2 with the definition of the energy efficiency metric and elements of the main elements required to compute it. In Sections 3, 4, 5, 6 and 7, we study respectively superconducting platforms, spin qubit platforms, trapped ions platforms, neutral atoms platform and photonic platform. In each of these sections, a baseline hardware component list and a compilation model are derived, after which the energy efficiency is showcased. Section 8 touches on error correction codes and future research directions opened by this work, and Section 9 concludes the work with general remarks. Finally, Appendix A contains the hardware component lists for each platform, Appendix B gives additional information on our compilation models and Appendix C discusses different chip technologies for photonic computing.

2 Methodology

In previous works [3, 4], the energetic footprint of quantum computers has been calculated by identifying the energy needed to perform the building block tasks of quantum algorithms: single qubit gates, measurements, heat evacuation etc. This approach can be valuable to understand the theoretical energetic demand of the individual operations taking place during computation, but it should be understood more as a lower bound for resource demand, rather than an accurate estimation that matches real hardware. In this work, we approach the issue from a hardware-aware perspective, by estimating the consumption of quantum computers using power consumption values from existing components and fully working setups. Although this method is less suitable to predict the scaling laws of future quantum computers, it serves our purpose of displaying the energetic demands of hardware that exists or is within reach of current NISQ technologies.

Our energetic model of quantum computing is based on two main parameters: energy consumption of the hardware and number of computations performed in a given time. We first note that the power consumption of state-of-the-art quantum computers does not change significantly whether the device is computing or not. In most architectures, either the hardware elements involved in performing computation cannot be turned on and off between operations, or the passive power demand is orders of magnitude larger compared to the elements that are only involved on the computing process. In addition, the consumption of these systems can not be, in general, broken down into isolated operations or actions that they perform on the qubits (such as the gates, or the measurements). For these reasons, our model has a more experimental, hardware-aware approach, similar to Yehia et al.’s work [5], where the consumption of several communication protocols for photonic circuits was estimated based on the power demand of real hardware.

Since power consumption remains roughly constant regardless of whether a quantum computer is actively performing computations, our figure of merit should account for how efficiently a given platform executes quantum algorithms. For instance, given two quantum computing platforms π_A and π_B , if π_A consumes 10 times more but is 100 times faster at performing tasks than π_B , one might prefer to use platform π_A . We thus estimate the energy consumed by the devices given a certain period of time rather

than for a certain task. This leads us to the following high level definition of the energy efficiency of a quantum computer:

$$EE_{\mathcal{A}}^{\pi} = \frac{\# \text{ Algorithms performed in time } t}{\text{Energy consumed in time } t}. \quad (1)$$

As demonstrated later in this section, this metric is a time-independent function that quantifies the cost of running algorithms on a quantum computing platform π , in computation per Joule. It is not strictly speaking an efficiency, since it is not dimensionless, but it benchmarks appropriately the energy used by quantum computers in terms of computation per joule. This approach can be used to compare the consumption of different platforms performing the same task, by accounting for the time a certain computer takes to perform the targeted task. Precise estimation of how long each platform needs to compute a given function is however a challenging endeavor, out of the scope of this article. In the remainder of this section, we elaborate on how the energy consumed and the number of algorithms performed in a given time are estimated.

It should be noted that, while the energy efficiency might allow to compare between quantum computing platforms, this is not the main point of this work. Each platform has their pros and their cons and should not be examined solely with one metric. We acknowledge that the heavy task of benchmarking quantum computers is a global effort, with, for example, DARPA recently launching a Quantum Benchmarking Initiative (QBI) [6] or the Quantum Energy Initiative (QEI) working group in the IEEE [7]. Their approach usually consists either in defining a *price-per-qubit* metric or to find a task that would allow a fair comparison between platforms. This study takes an alternative approach, showcasing a metric that attempts to capture the computational capabilities of a quantum computer, and that can readily be used on NISQ devices to analyze different architecture.

2.1 Hardware power consumption

The total power consumption of a given quantum computing platform π is simply given by the sum of the power consumption of each hardware element (classical and quantum) that it involves :

$$P^{\pi} = \sum_{i \in \mathcal{H}^{\pi}} P^i, \quad (2)$$

where $\mathcal{H}^{\pi}$ is the list of hardware elements included in π (for example lasers, vacuum chamber etc.) and P^i is the power consumption of each of them. By simply multiplying P^{π} by a time t , we can estimate the energy consumption of the platform during this time.

An interesting quantity of study is $E^{\pi}(0)$, the initial energy required to turn on the quantum computer π . This cost is usually quite large, since most quantum computing systems require some cool-down time for their cryogenics or the preparation of vacuum. Studying $E^{\pi}(0)$ allows, for example, to know how long a quantum computer needs to remain active to compensate the simple fact of turning it on. In current implementations of quantum computers, this energy is spent only once. The computer can then be left turned on for months. For this reason, the initialization energy is not included in our energy efficiency metric.

Ignoring the initialization cost, one can estimate the energy consumption of a computer running for a time t as

$$E^{\pi}(t) = t \cdot P^{\pi}. \quad (3)$$

However, as explained before, this is not enough to estimate the energy efficiency of quantum computers. For a given period of time, two devices may be able to perform very different numbers of quantum algorithms. In addition, for the same device, two algorithms can take various computing times.

2.2 Number of algorithms performed in a time period

The number of quantum algorithms $N^{\pi}(t)$ that a platform π can perform in a time t is a much more delicate quantity to estimate than the energy consumed during this time. It highly depends on the

connectivity between qubits, the gate implementation, the sampling overhead, and many other features that can vary depending on the quantum computing platform, and even between architectures that belong to the same qubit implementation technology. In addition, one can account for an extra-layer of complexity due to error correction or mitigation, which increases the number of qubits, gates, classical overhead, and the algorithm compilation time. We leave the discussion on error correction and fault tolerant quantum computing for Section 8.1.

In general, the number of times a platform π is able to perform an algorithm $\mathcal{A}$ in a given time t is given by:

$$N_{\mathcal{A}}^{\pi}(t) = \frac{t}{t_{\mathcal{A}}^{\pi}}, \quad (4)$$

where $t_{\mathcal{A}}^{\pi}$ is the execution time of $\mathcal{A}$. We thus consider partial executions of an algorithm as useful computing time. This means that for example, if $N_{\mathcal{A}}^{\pi}(24h) = 300.75$, then the computer π can complete the algorithm 300 times and execute 75% of it in 24h. Not considering partial execution can also be done by rounding down the the above quantity. In that case we would have that if $t < t_{\mathcal{A}}^{\pi}$, then the computer does not have time to perform $\mathcal{A}$ and $N_{\mathcal{A}}^{\pi}(t) = 0$. For readability in this paper, we will omit rounding down this quantity and consider partial execution as useful time.

Combining Equations 3 and 4 leads to the following time-independent equation for the energy efficiency of a quantum computer π running algorithm $\mathcal{A}$:

$$EE_{\mathcal{A}}^{\pi} = \frac{N_{\mathcal{A}}^{\pi}(t)}{E^{\pi}(t)} = \frac{1}{t_{\mathcal{A}}^{\pi} \cdot P^{\pi}}. \quad (5)$$

By varying $\mathcal{A}$, one can benchmark the energy consumption of a quantum computer for various use cases. In the remainder of this section, we explain the model we use to estimate $t_{\mathcal{A}}^{\pi}$, the time it takes to perform a quantum algorithm, taking into account the native gate set and other specifications of π . This study focuses on a circuit-based computation model, while we leave the discussion of other paradigms, such as quantum annealing, or measurement based quantum computing to Section 8.2.

2.2.1 Running time of a quantum algorithm

In this work, we make a distinction between a *quantum algorithm*, which is a high-level description of a task, and a *quantum circuit*, which corresponds to a translation of the task in terms of quantum gates, ready to be run on a specific quantum computing device. The process of adapting the sequence of quantum gates to be run on a device is the so-called compilation, which we discuss in the next section.

Since quantum computers give access to measurement outputs, *i.e.* samples from probability distributions, circuits have to be implemented a certain number of times N_{samples} to get sufficient statistics. This number of shots depends on the noise of the hardware used and on the specifics of the algorithm. Note that we use the terms sample and shot interchangeably. This leads to the following formula for the time necessary to perform a quantum algorithm $\mathcal{A}$ in a device π :

$$t_{\mathcal{A}}^{\pi} = t_{\text{circuit}}^{\pi} \cdot N_{\text{samples}}, \quad (6)$$

Throughout this work, we examine five quantum computing platforms that can be grouped in three families : *solid-state* devices, comprising superconducting and spin-based qubits, *atom-based* devices, which use neutral-atom and trapped-ion qubits and finally *photonic* devices. To factor in each platform's specificity, a different model to estimate the time to execute a circuit t_{circuit}^{π} is derived for each of these families.

In the case of solid-state platforms, we consider that a circuit can be broken down into different operations: reset of the qubits, gate application and then measurement. In general, a quantum circuit has a certain *depth* D , which represents the number of time-steps necessary to perform the circuit gates. We consider that synchronous programming languages are used to translate algorithms on quantum hardware. This means that the computer uses an internal clock to define time steps. Each of these

time-steps is composed of different basic operations that happen in parallel, namely implementation of single- and two-qubit gates. This leads us to the following formula:

$$t_{\text{circuit}}^{\text{solid-state}} = t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}}, \quad (7)$$

where t_{reset} is the time necessary to reset the qubit between every shot of the algorithm and t_{meas} is the time needed to read the final state of the qubits at the end of the circuit. The clock time t_{clock} corresponds to the time the computer requires to execute a single time-step. These three time parameters depend on several hardware parameters and are discussed in detail in the section pertaining to each platform.

In the case of atom-based platforms, the reasoning is similar but an additional component is considered in the estimation of the time to apply a circuit: atoms need to be moved to make use of their all-to-all connectivity. This is typically done by moving from an *storage zone* to different *gate zones*, in which gates can be applied. The execution of the circuit is thus similar to solid-state platforms, but an extra term accounts for atom transport between layers that require different sets of qubits. The formula we use is:

$$t_{\text{circuit}}^{\text{atom-based}} = t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}} + \beta_{\text{trans}} \cdot D \cdot t_{\text{trans}}, \quad (8)$$

where t_{trans} is the transport time and β_{trans} is the fraction of layers that requires changing the atoms that are held in the gate zones. This quantity highly depends on the structure of the circuits, and the number of gate zones on a given device, and we present results for different values of β_{trans} . We explain this in more detail in the appropriate sections and in Appendix B.2.

Lastly, photonic quantum computers operate in a totally different way. A circuit is not decomposed in layers of gates, but rather seen as a unitary that can be implemented in optical components arranged on a chip. Single photons are produced by a source, sent on the chip and measured using single photon detectors. Since the travel time of the photons from the source to the detector is negligible, the main parameters that quantify the time to perform a successful measurement on a photonic device are the rate of the source r_{source} and the end-to-end transmission probability η of the chip. We get that:

$$t_{\text{circuit}}^{\text{photonic}} = \frac{n}{r_{\text{source}}} \cdot \frac{1}{\eta^n}, \quad (9)$$

where n is the number of photons involved. The end-to-end transmission η depends on the optical depth, *i.e.* the number of layers of optical component on the chip. We explain it in more detail in the relevant section.

Beyond finding values for the relevant hardware parameters for each platform, the above equations present a challenge that is common to both solid-state and atom-based platforms. Namely: how can we compute the depth of a circuit implementing an algorithm on a specific platform?

2.2.2 Compilation of an algorithm into a quantum circuit

Compilation of a quantum algorithm into a hardware-aware quantum circuit is a complex optimization process. In photonic chip-based platforms, compilation consists in a single classical computation step which translates the algorithm into a configuration of the chip elements. The chip reconfiguration procedure is discussed in section 7. The rest of this section is relevant for compilation of algorithms executed on solid-state and atom-based platforms, where quantum circuits consists of D layers of gates. Gates can be arranged in multiple ways to optimize the overall depth, map logical operations to physical qubits, properly schedule operations etc. In this work, we simplify this compilation process in two steps: (1) *transpilation*, where the algorithm is translated into a circuit using the native gate set of the platform considered and (2) *routing*, where this circuit is adapted to the constraints of the platform, as could be, for example, the limited connectivity of superconducting chips or the movement of atoms to interaction zones in ion-based computers.

We first examine the effect of the transpilation step. Depending on the native gate set of a platform, less time steps might be necessary to perform an algorithm compared to another, even if they have access to the same number of qubits. Using Qiskit [8], we show in Fig. 1 how the set of quantum gates

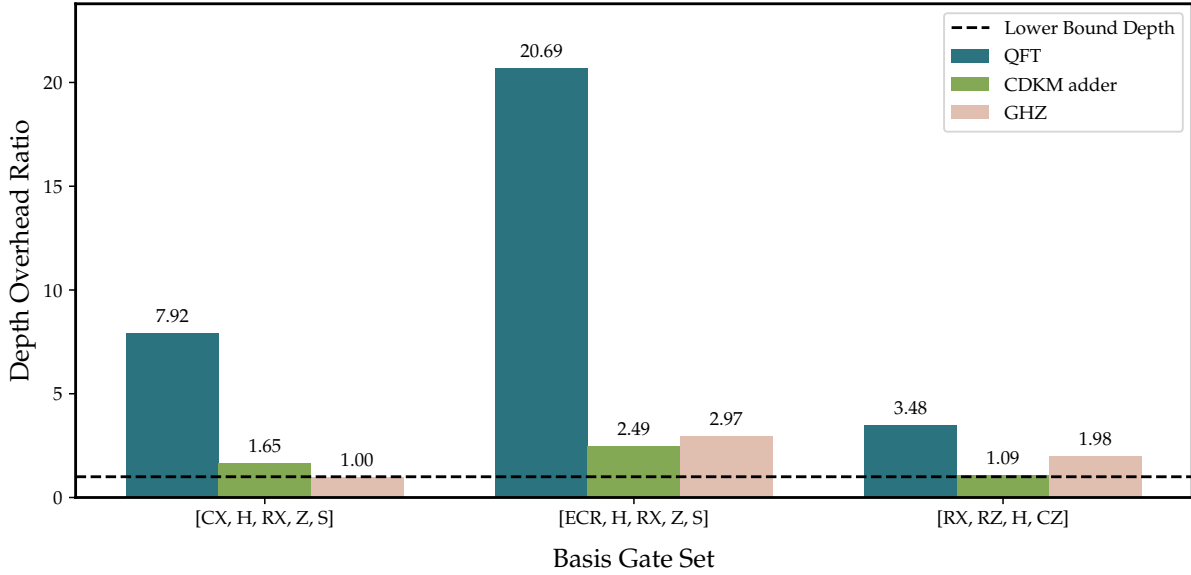


Figure 1: *Effect of the transpilation step.* Relative depth overhead ratio of the QFT, CDKM adder and GHZ state preparation circuits for different basis gate sets in a 100 qubit quantum computer, based on the Qiskit transpiler. The labels of the gates in the x-axis correspond to the Pauli Z gate (Z), controlled gates (CX and CZ), the Hadamard gate (H), parametric rotations (RX and RZ), the phase gate (S) and the maximally entangling gate (ECR). The lower bound (dashed line) is computed taking into account all the accessible gates in Qiskit.

that is accessible to a platform influences the final depth of a circuit for three different examples: the Quantum Fourier Transform (QFT), the CDKM adder [9] and an algorithm for $|\text{GHZ}\rangle = \frac{|0\dots 0\rangle + |1\dots 1\rangle}{\sqrt{2}}$ state preparation.

Once a gate set is fixed, the depth of a circuit is also influenced by the classical transpilation method used to translate an algorithm into these native gates. Very developed methods can result in a lower circuit depth at the cost of a longer classical computation time. There is thus a trade-off between the energy spent in compilation and the depth of the final circuit to be run on a quantum computer. In this work, we simplify this issue and consider that algorithms are pre-transpiled on classical machines and scheduled to run on the quantum computer. This corresponds to the cloud-based picture that is currently in place, where quantum devices are remotely accessible to end users. In this way, the time where the quantum machine is used is optimized. The transpilation time is assumed to be the same order of magnitude for all platforms, and that it is performed on classical computers in parallel to the quantum computations.

The routing step of the compilation consists in adapting the circuit to the specificity of the platform, ensuring that gates can be applied on the appropriate qubits. In the case of solid-state platforms, the main constraint is the connectivity between qubits. Qubits positioning in a superconducting and spin-qubit computers is usually modeled with a graph where nodes are the qubits and edges are the ability of two qubits to interact. Since two-qubit gates require that the qubits are able to interact, additional gates might be necessary to move qubits close together. This can be done using SWAP gates, that we consider here to be composed of 3 CNOT gates. As a result, each two-qubit gate implies a depth overhead on the post-transpilation depth D_0 , which depends on the distance between the relevant nodes in the processor. To have a general description of the overhead required for any given circuit, we introduce the parameter $\alpha_{2q} = \frac{D_0^{2q}}{D_0}$ which corresponds to the ratio of layers that include two-qubit gates D_0^{2q} over the total number of layers before routing. We approximate that the final depth of a circuit of transpiled

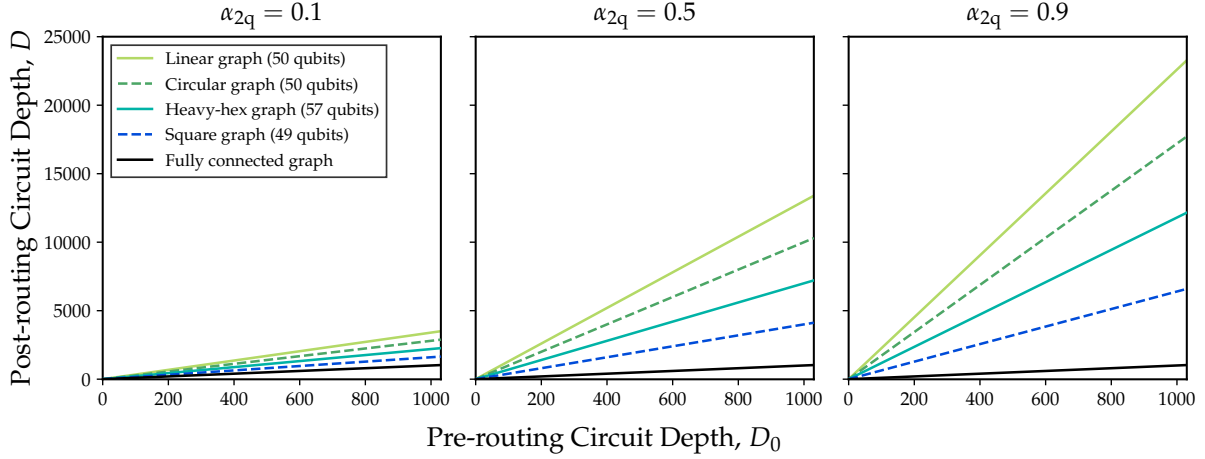


Figure 2: *Effect of the routing for solid-state platforms.* Post-routing circuit depth of a circuit as a function of D_0 . The three panels correspond to different ratios of layers with two-qubit gates α_{2q}

depth D_0 executed in a solid-state qubit based processor can be approximated as follows:

$$D^{\text{solid-state}} \approx D_0 + \alpha_{2q} \cdot 3 \left\lceil \frac{\overline{d(G)} - 1}{2} \right\rceil, \quad (10)$$

where $\overline{d(G)}$ is the average distance between nodes of the graph G representing the qubit connectivity in the platform. Since most quantum circuits are expected to include a high number of two-qubit gates, we can estimate the ratio α_{2q} to be close to 1 in most cases. Fig. 2 shows how the graph connectivity and the ratio of two-qubit gates influences the number of times a given circuit can be performed in 24 hours. In the case of all-to-all connectivity, modeled by a fully connected graph, no overhead would be induced by the two-qubit gates. All curves in this plot would bunch into the lighter green one. In Appendix B.1, we go in more detail in how these plots are computed.

In contrast, atom-based computers have all-to-all connectivity, but qubits need to be placed close to each other in order to interact. Therefore, gates are usually performed in interaction zones (or gate zones), where the qubits are brought together. The amount of interaction zones and the number of qubits they can host depend on the device considered. To address this constraint we consider two general scenarios: (a) the original circuit has less gates per layer than the available number of gate zones, in which case the depth of the circuit remains the same, and (b) on average, the number of gates per layer of the original circuit is higher than the number of gate zones. In this case, the maximum amount of gates performed per layer is limited by the number of zones. The depth of the circuit becomes:

$$D^{\text{atom-based}} = \left\lceil \frac{\overline{N_g^L}}{N_{\text{zones}}} \right\rceil \cdot D_0, \quad (11)$$

where $\overline{N_g^L} = N_g/D_0$ is the average number of gates per layer of the original circuit and N_{zones} the number of gate zones of a given device. Here, it is assumed that each gate zone can allocate 1 or 2 qubits, depending on the gate that is being performed. However, some devices can host multiple atoms in a single zone and perform gates on all of them simultaneously. In that case, N_{zones} is assumed to be equal to the number of atoms held in the entire zone. We go in more detail, including some special cases, in Appendix B.2.

Finally, it is worth to mention that there are a plethora of compilation techniques that can be used to compress the circuits further, and are not considered in this work. Research on optimized compilers for quantum algorithms is still ongoing. Some studies [10, 11] estimate that the depth of a circuit factorizing

a 2048-bit integer is $1.80 \cdot 10^{12}$. Quantum Fourier Transform on n qubits, which is a basic component of many quantum algorithms, typically has a depth of n , but approximations can be achieved in depth $\log(n)$ [12, 13]. In quantum chemistry applications, estimation of the depth of a circuit employed to elucidate reaction mechanisms in complex chemical systems vary in the range of 10^{10} and 10^{15} [14, 15]. While these numbers are generally very large, VQE applications seem to require circuit depths of the order of 100 [16]. This seems to be more in line with current quantum computer capabilities, whose qubit coherence time limits the depth to a few hundred layers. It is also worth noting that these values do not take into account the platform-dependent constraints that we mentioned in this section. There is a tradeoff to be considered between the resources devoted to the compilation of a circuit, and the resources that will be spent executing such circuit. For the sake of generality, we analyze the energy efficiency of quantum computers for generalized circuits that can be described by their depth, D . Therefore, we assume that the same algorithm can result in different circuits with a wide range of depth values, depending on the gate basis, connectivity, or any specific compilation techniques that might be used.

3 Superconducting qubits

Superconducting qubits are a solid state technology based on superconducting electrical circuits. They are also considered one of the best scaling technologies for quantum computing, thanks to the high addressability of each individual qubit in the chip and the compactness of semiconductor design [17]. For these reasons, this technology is being adopted and developed by some of the largest companies within the quantum computing industry, such as IBM [18], Google [19] and D-Wave [20].

3.1 Technology

By arranging capacitors, inductors, transmission lines, and other electronic components, one can create circuits with coherent quantized energy levels, that are analogous to qubits [21]. Any particular circuit can be associated to a Hamiltonian defined by its circuit components. The physical design and manufacturing of its main components results in the so-called characteristic energies. This will determine the anharmonicity and energy spectra of the qubits. Moreover, these qubits can be coupled between them in many ways (*e. g.* capacitively coupled or inductively coupled) modifying their Hamiltonian. This Hamiltonian development stemming from the design of the circuit is a step in the flourishing field of Circuit Quantum Electrodynamics (circuit QED).

The central quantum mechanical device that allows circuit properties to be used as qubits are the so-called *Josephson Junctions* (JJ) [22]. They consist of two superconductors separated by a very thin insulator (typically about 1nm thick). When held under an electrical current, the electrons grouped in Cooper pairs can tunnel from one superconductor to the other. JJ anharmonic quantum behavior is shared with the electrical properties of other components in the circuit, making them suitable to encode qubit states. There are three main superconducting qubit families: charge qubits, where the information is encoded in the number of cooper pairs on one side of the JJ; flux qubits, where the information is encoded in the direction of the current; and phase qubits, where the information is encoded in the phase difference across a JJ [23, 24]. The most common qubit design is the transmon qubit [25, 26], which is a charge qubit with a higher tolerance to charge noise. In this work, we consider a flux qubit architecture, the fluxonium [27, 28], adopted by Qilimanjaro.

In general, superconducting qubits are controlled with Rabi oscillations produced in the form of microwave oscillations [25, 26, 23, 24]. In a quantum chip, superconducting qubits would be interconnected within the same electrical circuit. By tuning the frequencies [26] of multiple qubits, they can become resonant and interact [23, 24, 29]. Typically, the measurement of a superconducting qubit state is done by dispersive readout. This technique consists in measuring the transmission coefficient of a resonator coupled to the qubit. It is important to note however, that the control and readout schemes of superconducting qubits might vary depending on the qubit design.

Superconducting qubits have to be typically kept at $T_q = 35$ mK or lower, to reduce the thermal noise that affects the fidelity of their quantum state [30] and maintain the metal component plates of

the Josephson junctions at superconducting state. This condition implies the use of powerful dilution refrigerators capable of reaching millikelvin temperatures [31] and evacuating the heat dissipated by the quantum chip and electronics. These refrigerators are known to be the a major factor on the overall energy consumption of superconducting computers.

In summary, superconducting computers are electrical circuits controlled by a series of microwave signals, that have to be held under extremely low temperatures. This setup requires a dilution fridge able to cool down the chip and the wires, which are connected to the control electronics that generate and send the microwave pulses (generally at room temperature).

3.2 Energy model

In this section, we break down the hardware components of a superconducting quantum computer and their corresponding consumption. In order to estimate its energy efficiency, we also list the main operations that define the time that this computer takes to execute a circuit. Both the hardware components and operation times are listed in detail in Appendix A.1.

3.2.1 Power consumption of the hardware

The components displayed in Table 1 of Appendix A.1 correspond to a superconducting-qubit computer that contains around 50 qubits connected in a 2D square lattice. Depending on the qubit architecture, a single feed line can power different numbers of qubits. In this case, we use Qilimanjaro’s fluxonium architecture [27, 28] as a reference. This reference model is intentionally simplified and serves as a baseline architecture for our estimations, rather than a description of a specific deployed system. To complement this reference architecture with a real-world validation, in the Appendix, we consider the experimentally characterized superconducting quantum processor QMIO [32], deployed in the Galicia Supercomputing Center (CESGA). According to the components shown in Table 1, the consumption of a superconducting-qubit computer can be estimated to be $P^{\text{supercond}} \approx 20$ kW. This corresponds to ≈ 1700 MJ of energy consumed in 24h.

The consumption contribution of the different components in the hardware is shown in Figure 3. In this figure, the hardware components are grouped according in three different areas: cooling, qubit control and classical processing.

As one can see, the highest power-demanding parts of the computer are devoted to the cooling system, with 80% (16.1 kW) of the overall consumption. It is unclear how the power of the refrigerator might scale with the number of qubits, but we consider this value representative to the superconducting devices of this scale (~ 50 qubits). As we have seen for almost every platform in this work, there is a common trend on consuming most power to produce the suitable physical environment to host qubits. As we mention above, superconducting qubits need to be held under vacuum and cryogenic conditions. However, we do not account for the power consumption of the vacuum pump in this analysis, since it is only used to initialize the computer, and its power consumption during computations is negligible.

There is a clearer dependency of the electronics needed in the readout lines with respect to the number of qubits. In particular, each readout line need its own radio-frequency (RF) source as well as two low-noise amplifiers (LNA). Considering each line can feed up to 5 superconducting qubits, we can compute the number of control qubits components for a given number of qubits. The control and readout cluster has a max power demand of 1.1 kW, and can be modified combining different modules that addresses the different signals needed for qubit control and readout. Therefore, its power consumption also depends on the number of qubits in the computer, and the lines required according to their architecture. For the sake of generality, we include its maximum consumption to our estimations. Finally, this component could be replaced by a combination of FPGAs. This is the case of the QMIO computer shown in Table 3, where each qubit has a dedicated FPGA to manage its signals. Compared to this particular FPGA design, the cluster turns out to be the most compact and efficient choice.

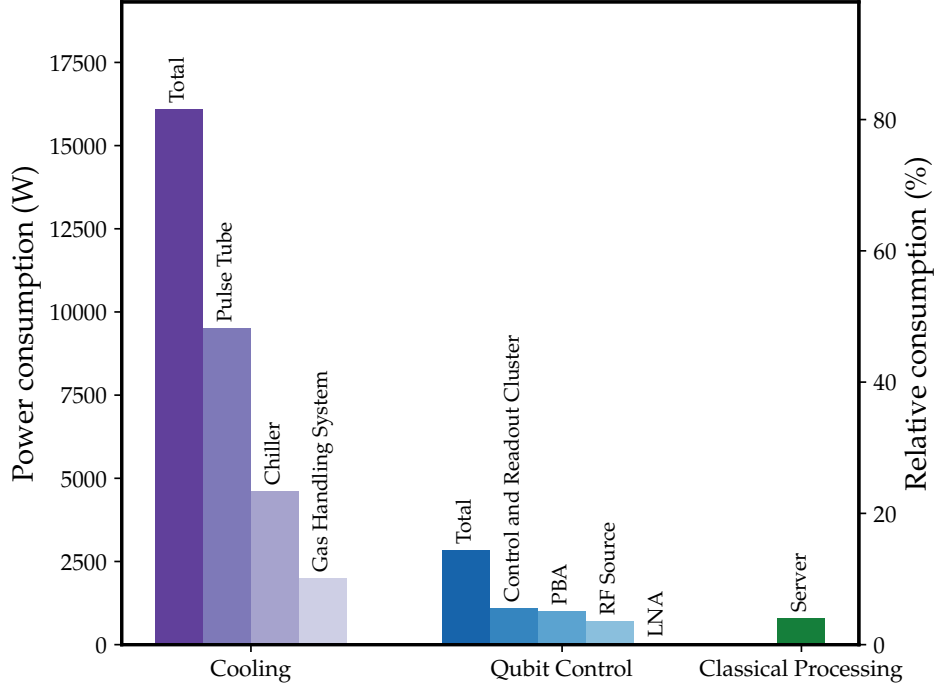


Figure 3: Power breakdown of a superconducting-qubit computer with ~ 50 qubits, according to the components shown in Table 1. In the x-axis, the components are grouped depending of their purpose in the computer. Each group has a bar that corresponds to the total accumulated power of its components. The left y-axis shows the absolute value of power consumption of the components, and the right y-axis the relative value with respect to the overall consumption of the computer.

3.2.2 Derivation of the energy efficiency

To compute the energy efficiency of running an algorithm $\mathcal{A}$ on a superconducting qubit platform, one first needs to compile $\mathcal{A}$ into a quantum circuit of depth D . As explained in the introduction and developed in Appendix B.1, we estimate the gate overhead due to the connectivity constraints of the platform by using the average distance between the qubits in the connectivity graph of the platform. We consider here that D is the final depth of a circuit implementing $\mathcal{A}$, which has been already compiled according to the gate set and connectivity of the device. By taking D as a parameter, the energy efficiency of the computer can be analyzed without having to define a specific task $\mathcal{A}$. The model of the execution time of $\mathcal{A}$ compiled into quantum circuit of depth D on a superconducting-qubit quantum computer is given by:

$$t_{\mathcal{A}}^{\text{supercond}} = N_{\text{samples}} \cdot t_{\text{circuit}}^{\text{supercond}} = N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}}), \quad (12)$$

where N_{samples} is the number of shots, t_{reset} is the time necessary to reset the qubit between every shot of the algorithm, t_{meas} is the time needed to read the final state of the qubits at the end of the circuit and t_{clock} is the clock time of the computer. Based on observations of current superconducting qubit devices, we define the following operation times: $t_{\text{reset}} \approx 5 \mu\text{s}$, $t_{\text{clock}} \approx 50 \text{ ns}$ and $t_{\text{meas}} \approx 1.6 \mu\text{s}$, which are further discussed in the Appendix A.1.1. This reset time corresponds to the active reset technology, which we compare with a passive reset time ($t_{\text{reset}}^{\text{pass}} \approx 200 \mu\text{s}$) at the end of the section; until then, we assume active reset.

Due to the dependency of t_{circuit} with the post-compilation depth D , it is important to note that the connectivity constraints of the processor will have a deep effect on the computing time, and therefore the efficiency of superconducting computers. In this section we do not analyze the efficiency of processors

with different graph connectivities, but one can see how the depth overhead scales with the number of qubits for different connectivity architectures in Appendix B.1. In addition Figure 2 shows values of the post-compilation depth D for different densities of two-qubit gates and connectivity graphs, for processors that have ~ 50 qubits.

The energy efficiency of a superconducting-qubit quantum computer is thus given by

$$EE^{\text{supercond}} = \frac{1}{N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}}) \cdot P^{\text{supercond}}}, \quad (13)$$

where $P^{\text{supercond}}$ is the sum of the power consumption of the hardware elements of Table 1.

3.3 Results and discussions

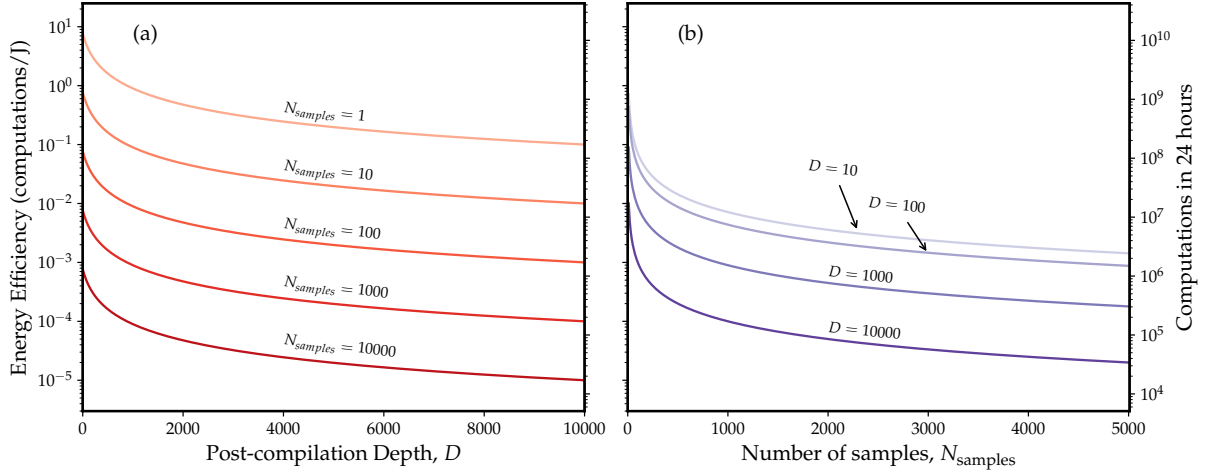


Figure 4: Energy efficiency (left y-axis) and number of computations (right y-axis) in 24 hours of a superconducting-qubit computer depending final circuit depth, D and number of samples, N_{samples} . Panel (a) contains several series with different N_{samples} values and the x-axis corresponds to D . Series in panel (b) have different D values and its x-axis corresponds to N_{samples} .

Figure 4, shows the energy efficiency and number of computations of a superconducting computer executing an algorithm with given D and N_{samples} for 24 hours. The relation between the energy efficiency and the number of samples is inversely proportional, according to Eq. 13. For this reason, the series shown in panel (a) are equidistant in the logarithmic scale, separated exactly by an order of magnitude.

One can see that the dependency of EE with the post compilation depth D presents a different behaviour. In panel (b), the series corresponding to $D = 10$ and 100 are closer together than the rest. This is due to the effect of the measurement and reset times at lower depths. For lower values of D , t_{reset} and t_{meas} are several orders of magnitude larger than t_{clock} , and therefore, they take most of the computing time. As an example, a sample of a $D = 10$ layers circuit would take $t_{\text{circuit}} = 7.1 \mu\text{s}$, which does not increase substantially at $D = 100$ layers, for which $t_{\text{circuit}} = 11.6 \mu\text{s}$. In both instances, most of the time is spent measuring and resetting the qubits ($6.6 \mu\text{s}$). For $D = 1000$ and 10000 layers, the clock time becomes more relevant; $t_{\text{circuit}} = 56.6$ and $506.6 \mu\text{s}$, respectively.

To analyze further the relation between t_{clock} , t_{reset} and t_{meas} , we compare the efficiency between computers with active and passive qubit reset. Figure 5 displays the energy efficiency of the computer with active reset ($t_{\text{reset}}^{\text{act}} = 5 \mu\text{s}$), passive reset ($t_{\text{reset}}^{\text{pass}} = 200 \mu\text{s}$), and instantaneous reset of the qubits ($t_{\text{reset}} = 0 \text{ s}$). As one can see, the efficiencies of the computer with passive reset display a lower and flatter profile than the active one. This is due to the high impact of the passive reset time during computation, which forces short circuits to take a very similar computing time. On the contrary, the active reset favors

the execution of lighter circuits. In particular, for short circuits of depth $D \approx 100$ layers, the device with passive reset has an efficiency of $EE = 2.5 \cdot 10^{-3}$ computations/J, 20 times lower than its counterpart.

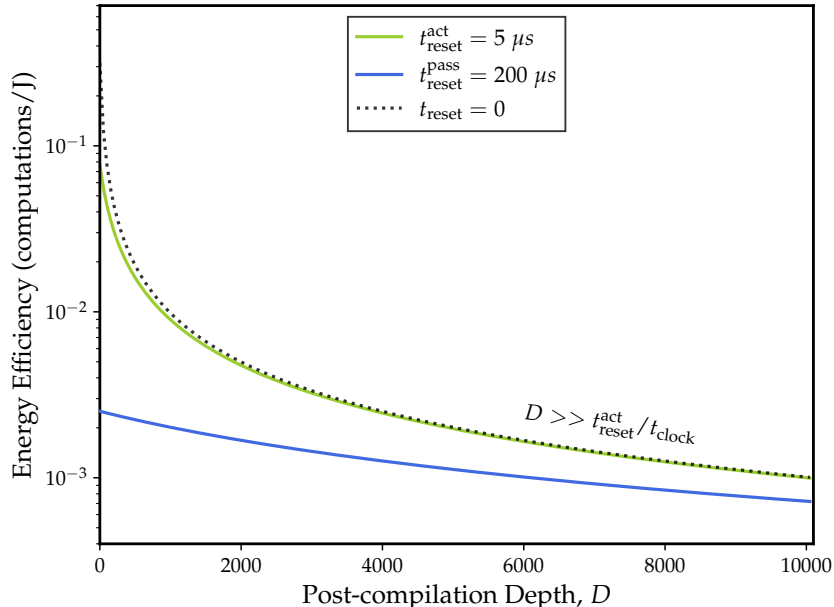


Figure 5: Energy efficiency of a superconducting-qubit computer, depending on the depth of the executed circuit and the reset time. The green line corresponds to a computer with active reset, the blue line to passive reset, and the dotted line to instantaneous reset.

In accordance with Eq. 13, Figure 5 shows how the efficiency becomes inversely proportional to D as the depth of the circuits increase, independently of t_{reset} . This happens when the circuit is deep enough ($D \gg \frac{t_{\text{reset}} + t_{\text{meas}}}{t_{\text{clock}}}$) for the time taken to measure and initialize the qubits to be negligible in comparison with the operation time $D \cdot t_{\text{clock}}$. In that regime, the energy efficiency curve converges to the black dotted curve that represents an ideal computer with no reset time. In the case of the computer with active reset, this can be already seen at $D \sim 10^3$. However, for the computer with passive reset this regime is shifted to higher circuit depths ($\sim 10^5$ layers).

Superconducting qubits come in a wide variety of designs. The choice of qubit architecture and chip topology, together with the type of coupler used (whether tunable or fixed), shapes the effective Hamiltonian of the system and directly influences its anharmonicity, decoherence rates, and susceptibility to noise. Beyond the physics, these design choices also determine the number of drive and readout lines required, which in turn sets the scale of the energy needed for control and readout operations. As a general rule, each loop in the superconducting circuit requires its own drive line. A standard transmon, for instance, contains a single loop and therefore needs only one drive line. However, architectures that incorporate a Superconducting Quantum Interference Device (SQUID) introduce an additional loop, meaning that a single qubit in such a design would require two drive lines rather than one.

This inherent diversity in superconducting qubit design makes it remarkably difficult to produce a unified estimate of energy consumption and efficiency across the full landscape of architectures. Parameters such as native gate sets, number of loops, coupler type, and resulting qubit connectivity all vary significantly from one design to another, and each of these factors influences how efficiently a given algorithm can be executed. While the individual components making up these architectures are generally expected to behave homogeneously, the specifics of circuit depth required for a particular algorithm implementation, the number of drive lines, and how both of these scale with increasing qubit count are highly architecture-dependent and far from straightforward to generalize.

4 Silicon spin qubits

Loss and DiVincenzo proposed a spin-based approach to quantum computation by using the spin of trapped charges, such as electrons confined in quantum dots, to encode quantum information [33]. This approach exploits the spin- $\frac{1}{2}$ nature of electrons, which inherently provides a two-level quantum system. Leveraging complementary metal-oxide-semiconductor (CMOS) technologies, spin qubits are a strong candidate for scaling up future quantum computers. Recent work has demonstrated key building blocks for quantum error correction with silicon spin qubits, such as weight-four parity checks, though these implementations rely on more complex architectures involving electron shuttling [34]. Spin qubits are being investigated by many companies (Intel) and startups (Quantum Motion, Diraq, Quobly, Equal1 and many more).

4.1 Technology

Several physical platforms have been developed to realize spin qubits, each relying on a different way to confine and manipulate individual spins. The first proposal, introduced by Loss and DiVincenzo in 1998 [33], encodes quantum information in the spin state of a single charge carrier—either an electron or a hole—trapped inside a gate-defined quantum dot in a semiconductor. In this configuration, the qubit logical states correspond to the spin-up and spin-down projections of that confined particle.

Alternative implementations exist. One approach uses a donor atom embedded in a semiconductor lattice, where the electron remains bound to the impurity potential [35]. Other designs exploit the collective spin states of two or more electrons confined in coupled quantum dots: for instance, the singlet-triplet qubit in a double quantum dot [36], or the exchange-only qubit realized in a triple quantum dot [37]. These multi-spin encodings can provide some resilience to charge or magnetic noise, at the cost of more complex control schemes. This study restrains itself exclusively to single-spin qubits confined in gate-defined quantum dots, as this platform currently offers the most mature technology for semiconductor-based quantum computing. Spin-qubit arrays with up to twelve [38] and ten [39] qubits in silicon and germanium quantum wells, respectively, were demonstrated. In parallel, single- and two-qubit gate fidelities have overcome the fault-tolerance threshold for quantum error correction [40, 41, 42, 43, 44].

A magnetic field lifts the degeneracy between the spin-up and spin-down states through the Zeeman effect, defining the qubit energy splitting. Spin rotations (i.e. single-qubit gates) can be driven electrically by taking advantage of spin-orbit coupling. This coupling can be intrinsic, as is the case for holes in silicon or germanium, where the spin and orbital motion are naturally entangled [45], or engineered by introducing a magnetic field gradient using on-chip micromagnets, enabling electrical control of electron spins in otherwise weakly spin-orbit-coupled materials [46].

Two-qubit gates are typically realized through the exchange interaction, which arises when the wavefunctions of neighboring electrons overlap. Because of the Pauli exclusion principle, electrons with parallel spins experience a higher energy cost than those with opposite spins, effectively producing a spin-spin coupling that can be tuned by gate voltages [47]. Beyond nearest-neighbor coupling, long-range two-qubit interactions have recently been demonstrated using photon-mediated coupling through superconducting microwave cavities [48], opening the door to modular spin-based quantum processors.

The state of a spin qubit is read out indirectly by converting spin information into charge information. One of the most common methods relies on Pauli spin blockade: two electrons with the same spin cannot occupy the same quantum dot orbital, even if it would minimize the electrostatic energy [49]. This spin-dependent charge configuration can then be detected by a nearby charge sensor, such as a quantum point contact or a single-electron transistor. An alternative approach is RF gate-based sensing, where the charge configuration is inferred from the change in impedance of an RF resonant circuit, which can be either classical [50] or superconducting [51].

Spin qubit devices are typically operated at millikelvin temperatures in a dilution refrigerator, comparable to the conditions required for superconducting qubits. Operating at such low temperatures minimizes thermal excitations that could otherwise randomize the spin state. Remarkably, recent experiments have shown that coherent spin-qubit operation remains feasible up to 4 K [52], which would enable simpler and more compact cryogenic setups without the need for ^3He -based refrigeration. Similarly to

superconducting quantum processors, spin-based chips are mounted on the coldest plate of the dilution refrigerator. Electrical control and readout signals are delivered through DC and coaxial lines connecting the cryogenic environment to the electronic control hardware at room temperature.

4.2 Energy model

This section proceeds to discuss the hardware elements present in a spin qubit quantum computer, and the operation times required to execute circuits. Due to the current state of spin qubit development, the present devices fall short in terms of number of qubits in comparison with the other platforms. For that reason, the estimations shown in this section are calculated by extrapolating the number of hardware components that the current devices require to a larger amount of qubits, to provide estimations that correspond to a computer of similar size to those shown in other sections. In addition, we assume that the cryogenic setup for a spin qubit computer is equal to the one corresponding to a superconducting computer of the same size, presented in the previous section. Appendix A.2 contains the detailed list of components and operation times of the considered spin qubits processor design.

4.2.1 Power consumption of the hardware

The hardware components of the baseline spin-qubit computer are shown in Tables 4 and 5 of Appendix A.2. In this case, the number of components corresponds to a 7×7 2D square lattice architecture, with $N_q = 49$ qubits. We consider a full one-to-one control scheme where each qubit is addressed with an individual driving line. Resonators are placed on half of the qubit sites to perform parity readout. The number of resonators in this architecture would therefore be $N_R = \lceil N_q/2 \rceil = 25$. The end of this section discusses the energy efficiency of different architectures with more complex qubit connectivity based on multiplexing.

Taking the components shown in Table 4, a spin qubits quantum computer with $N_q = 49$ qubits, would consume $P^{\text{spin}} \approx 20$ kW of power. Therefore, having this computer turned on for a day would cost ≈ 1700 MJ of energy. Figure 6 shows the power demand of all the components in the computer, split in three groups: cooling, qubit control and classical processing. As one can see, the power consumption is distributed similarly to the superconducting computer examined in Sec. 3.3. Likewise, the cooling hardware of a spin-qubit computer consumes 16.1 kW of power, which corresponds to the 81% of the total consumption of the device.

As we mention above, we consider the signals that control the qubits to be addressed individually by dedicated electronics. Then, the number of lines does not only depend on the number of qubits, but also the connectivity of the processor. In the following section we explore the energy efficiency of different architectures, based on their connectivity graph and their multiplexing capabilities. Although the increase in the power consumption of qubit control electronics can be clearly analyzed, it is unclear how increasing the number of lines and qubits will affect the power consumption of the cryogenic setup.

4.2.2 Derivation of the energy efficiency

The derivation of the energy efficiency of running an algorithm $\mathcal{A}$ on a spin qubit platform is identical to the one for the superconducting qubit platform presented in Section 3.2.2. Given the final depth D of a circuit implementing $\mathcal{A}$ on a spin qubit quantum computer, the execution time of $\mathcal{A}$ is given by:

$$t_{\mathcal{A}}^{\text{spin-qubit}} = N_{\text{samples}} \cdot t_{\text{circuit}}^{\text{spin-qubit}} = N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}}), \quad (14)$$

where N_{samples} is the number of shots, t_{reset} is the time necessary to reset the qubit between every shot of the algorithm, t_{meas} is the time needed to read the final state of the qubits at the end of the circuit and t_{clock} is the clock time of the computer. In order to provide estimations, we base these values on the updated review of spin-qubit computers by Stano and Loss [53]. According to their survey, these operation times can vary several orders of magnitude, even for spin qubits of the same species. In this work, we consider the following average values: $t_{\text{reset}} \approx 0.1$ ms, $t_{\text{meas}} \approx 10$ μ s and $t_{\text{clock}} = t_{2q} \approx 10$ ns. Comparing these operation times, we expect to see the computations in spin-qubit computers to be mainly

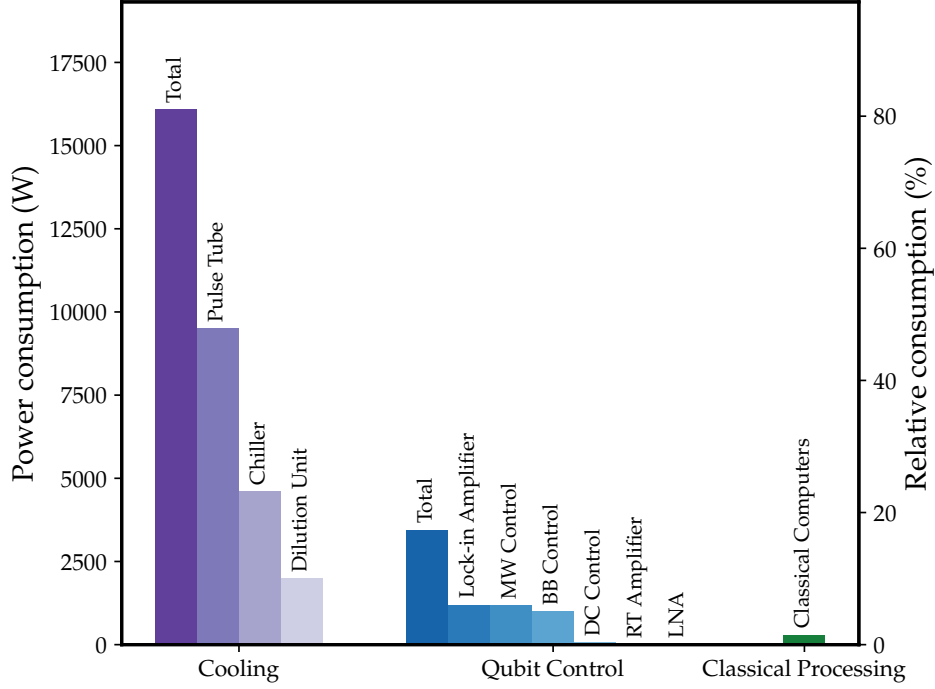


Figure 6: Power breakdown of a spin-qubit computer with 49 qubits, according to the components shown in Table 1. In the x-axis, the components are grouped depending of their purpose in the computer. Each group has a bar that corresponds to the total accumulated power of its components. The left y-axis shows the absolute value of power consumption of the components, and the right y-axis the relative value with respect to the overall consumption of the computer.

determined by the reset and measurement time of the qubits, which are several orders of magnitude larger than the clock time. The energy efficiency of a spin-based quantum computer is thus given by

$$EE^{\text{spin}} = \frac{1}{N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}}) \cdot P^{\text{spin}}}, \quad (15)$$

where P^{spin} is the sum of the power consumption of the hardware elements of Table 4.

4.3 Results and discussion

In Figure 7, we show the energy efficiency (left y-axis) and number of computations (right y-axis) of a spin-qubit computer executing a circuit with given D and N_{samples} for 24 hours. Since EE^{spin} and N^{spin} are inversely proportional to the number of samples, N_{samples} , the different series in panel (a) are equidistant on a logarithmic scale, shifted by an order of magnitude.

Interestingly, the EE curve profiles appear almost flat, in contrast with the results discussed in the superconducting qubits section 3.3. This behavior is directly related to the relatively high initialization time of spin qubits. As mentioned above, the clock time does not play any major role until the number of layers is large enough. Therefore, one can see how the energy efficiency of the device barely decreases as the depth of the circuit grows. As an example, running a single shot of a circuit with $D = 100$ layers would take $t_{\text{circuit}} = 111 \mu\text{s}$, where reset and measurement operations took 99% of the time. However, if we increase the depth of the circuit enough ($D \approx t_{\text{reset}}/t_{\text{clock}} = 10000$), the clock time starts taking over the majority of computing time. To put this into perspective, a single sample of an extremely long circuit with $D = 10000$ layers would take $t_{\text{sample}} = 210 \mu\text{s}$. In this case, the spin-qubit device would spend almost half of the computing time initializing the qubits.

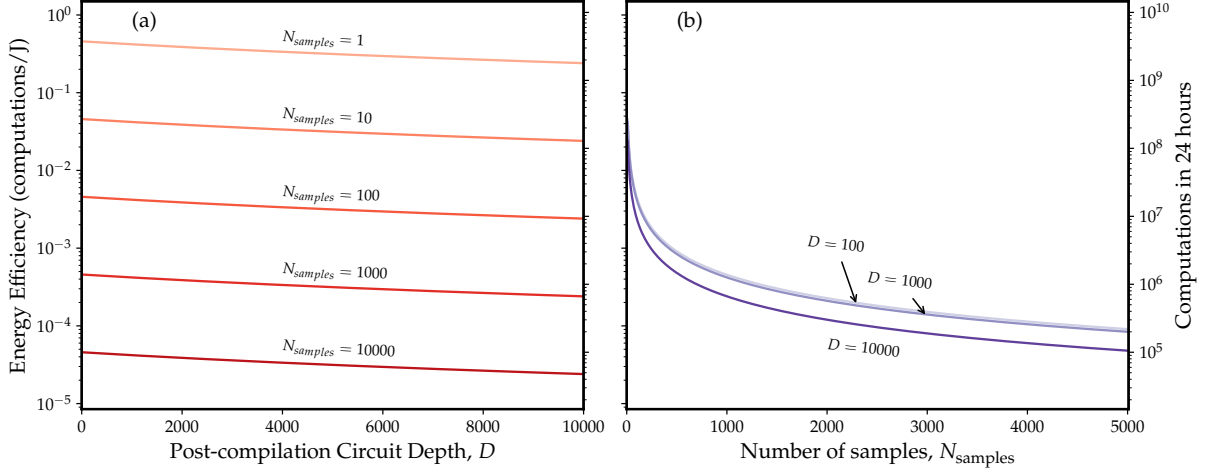


Figure 7: Energy efficiency (left y-axis) and number of computations (right y-axis) in 24 hours of a spin-qubit computer depending on final circuit depth, D and number of samples, N_{samples} . Panel (a) shows several series with different N_{samples} values and the x-axis corresponds to D . Series in panel (b) have different D values and its x-axis corresponds to N_{samples} .

With that in mind, it is clear that near term machines that will run short circuits can improve their energy efficiency by optimizing the initialization time of the qubits. However, this improvement would only be perceived for circuits with a relatively short depth ($D < t_{\text{reset}}/t_{\text{clock}}$). A representation of this behavior can be seen in Figure 5, where decreasing the initialization time of superconducting qubits leads to pronounced gradients of the EE with respect to D , for relatively short circuits.

The panel (b) of Figure 7 displays the values of EE and N^π with respect to the number of samples, N_{samples} , taken for three different circuits with depths $D = 100$, 1000 and 10000 . Here, the gap between lines is also determined by the initialization time of the qubits, for the same reason we explained above: the clock time is not the main contribution of the computing time, until the depth of the circuit is the order of $D \approx 10000$ layers. Therefore, the $D = 100$ and $D = 1000$ series in panel (b) have almost the same values, since these circuits take almost the same time to be computed.

Additionally, two extra features are analyzed, that have a major impact on the computation time and energy expenditure. Until now, spin qubits processors with a 2D qubit connectivity have been considered. However, due to the small number of qubits in actual processors, current devices are often displayed with linear graphs [53]. We proceed to demonstrate how the energy efficiency of spin-qubit computers would scale if they were developed according to different qubit connectivities rather than a two-dimensional one. Section 2.2.2 have already shown the poor scaling of linear graphs, in comparison with higher-dimensional ones. The same comparison can be seen here, applied to the specific case of spin qubits components and operation times.

Figure 8 shows the energy efficiency as a function of the pre-routing depth of the circuit, D_0 , for a ratio of layers including two-qubit gates $\alpha_{2q} = 0.5$. This value corresponds to a circuit that has at least one two-qubit gate on half of their layers, after transpilation. The figure includes values corresponding to linear (solid light green) and 2D (solid dark green) connectivities. In addition, we include the energy efficiencies of a processor with a "Crossbar" architecture, that is discussed at the end of the section.

For this comparison, we consider that the computers with linear and 2D connectivities share the majority of components, besides the number of DC control lines and base-band (BB) control lines. Since the spins in the linear processor have fewer nearest neighbors than the 2D processor, they also have fewer coupling signals, thus having reduced numbers of DC and BB lines. In Table 4 and 5, we break down all the number of lines for each computer.

Due to this line reduction, the power consumption of the linear processor saves 200 W of power. Despite this, the linear processor quickly becomes less efficient than the 2D one as we increase the pre-

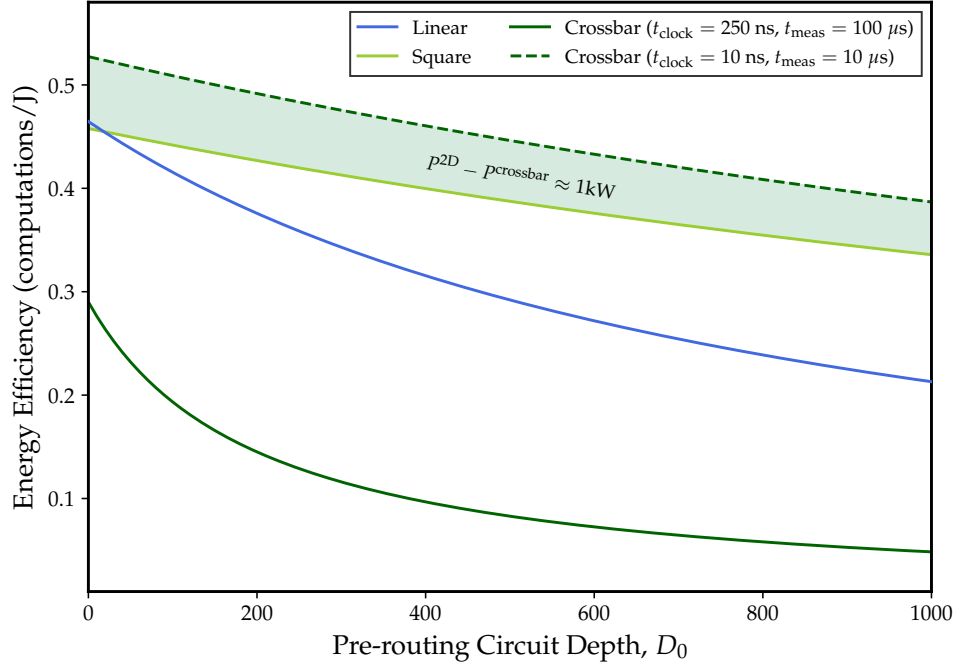


Figure 8: Energy efficiency of a spin-qubit computer executing 1 sample of a circuit, depending on its pre-routing depth D_0 . The blue curve corresponds to a computer with a linear connectivity graph, the solid light green to a 2D square-lattice connectivity graph, the dark green to a crossbar device with computational times taken from Ref. [54], and the dashed dark green line to a crossbar device with the computational times detailed in Sec. 4.2.2. The shaded area marks the difference in EE between the 2D processor and a Crossbar architecture that share the same operation times, thanks to the reduced number of lines. To compute the final circuit depth of the compiled circuits, we assume a ratio of layers including two-qubit gates $\alpha_{2q} = 0.5$.

routing depth of the executed circuit. Such efficiency loss is due to the extra-computing time that the processor needs to execute a larger amount of SWAP gates. It should be noted that, for a given D , both processors would take the same time and consequently, consume the same amount of energy. In order to interpret Figure 8 correctly, one has to consider that for the same value of pre-routing depth, D_0 , the linear processor has a larger SWAP gate overhead than the 2D one, and thus executes the same circuit with a larger final depth D .

With that in mind, it is clear how the large SWAP overhead due to the limited linear connectivity results in a clear energy efficiency drop. To put things in perspective, for a precompiled depth of $D_0 = 500$ and $\alpha_{2q} = 0.5$ the final depth of the circuit for the 2D-connected processor is $D^{2D} = 2000$. Meanwhile, the final depth of the same circuit for a linearly-connected processor is $D^{\text{linear}} = 6500$. The computation time of these circuits are $t_{\text{circuit}}^{2D} = 0.13$ ms and $t_{\text{circuit}}^{\text{linear}} = 0.17$ ms. This is another example on how the initialization time is still relevant even for circuits that are far apart in terms of depth. We estimate the energy efficiency of these devices for such circuits to be $EE^{2D}(D_0 = 500, \alpha_{2q} = 0.5) = 0.387$ computations/J and $EE^{\text{linear}}(D_0 = 500, \alpha_{2q} = 0.5) = 0.292$ computations/J.

Finally, we extend the previous comparison to an additional architecture proposed by Li *et al.* in Ref [54]. In this architecture, a 2D lattice of quantum dots is controlled using superconducting strips that carry multiple control signals among rows, columns and diagonals of qubits. All the signals controlled by each bar can be multiplexed into a single wire, reducing the total number of lines and their corresponding active electronic components. With these multiplexed signals, we estimate that a 49 spins processor could save up to ≈ 1 kW of power, thanks to the reduction of lines and control electronics. This estimation, however, only takes into account the power directly delivered to the hardware that we removed from the

previous non-multiplexed architecture, but does not consider the potential reduction in terms of cryogenic demands thanks to fewer sources of heat dissipation inside the cryostat. In Appendix A.2 we include more details regarding the number of lines, electronics, and operation times of the crossbar architecture.

Figure 8 shows the energy efficiency as a function of the pre-routing circuit depth D_0 , assuming it has a 2D connectivity graph. First, we estimate the EE of a crossbar processor based on the clock time $t_{\text{clock}}^{\text{crossbar}} = t_{\text{gate}} \approx 250$ ns given in the original article. Additionally, we assume that the measurement time of this device would scale with $\sqrt{2N_q}$, since the corresponding signals of entire rows of qubits should be time multiplexed in to individual wires, hence, $t_{\text{meas}}^{\text{crossbar}} \approx \sqrt{2N_q} \times 500$ μs . With these numbers in mind, one can see how the energy efficiency of such device (dark green dashed line) would drop drastically as the pre-routing depth of the circuits increase, in comparison with the previous architectures with a clock 25 times faster. Interestingly, we see how the linearly-connected processor outspeeds the crossbar one, even with its constrained connectivity, which leads to higher efficiencies. This result stresses the importance of computational speed for energy efficiency, as given by Eq 1.

Additionally, Figure 8 shows the same profile for a crossbar processor that has the same operation times as the Linear and 2D processors (dotted dark green line). It is unclear whether such times would be achievable with this architecture, but this result shows how, even with a substantial power reduction of ≈ 1 kW of saved power, computational times can have a larger effect on the overall efficiency of the computer.

5 Trapped Ions

Trapped ions [55, 56, 57], are an example of a different kind of quantum computer with respect to the previous sections, based on entangling the electronic state of different atoms through controlled collisions. That is to say, the electronic states of different atoms are correlated by controlling the centre-of-mass degrees of freedom to modulate the interaction between the internal states [58, 59]. In addition, the internal electronic states of individual ions can be separately controlled and measured. Atom-based platforms like trapped ions, or neutral atoms, proved to be a feasible alternative to build quantum computers at room temperature. Some of the main adopters of trapped ion quantum computers are IonQ [60] and Quantinuum [61].

5.1 Technology

Trapped ion computers consist of a chain of ionized atoms that are trapped using time-dependent electromagnetic fields. State preparation and gate drive is performed with lasers and/or microwave fields. Each ion in the chain behaves as a physical qubit, and their internal energy states are used to encode information. Typically, those are internal electronic states, corresponding to those of an uncoupled, valence electron (known as optical levels), those of the electron coupled with the atom nucleus (known as hyperfine levels) [56], or the electronic energy states on a magnetic field (Zeeman states). High fidelity gates have been demonstrated for different choices of states [62, 63, 64]. Laser cooling is used to reduce the kinetic energy of the ions so that motional dephasing is suppressed. On top of that, a ultra-high vacuum environment is required, as background collisions are the main driver of atom loss.

Single- and two-qubit gates can be implemented by irradiating the ions with electromagnetic field pulses, either using lasers or microwaves. For single qubit gates, the driving field only changes the internal electronic states while for two qubit gates the internal electronic states and centre-of-mass motional states are coupled, requiring 'phonons' to mediate the interaction as well [57, 56]. Movement of the ions can be achieved by changing the electromagnetic potential that confines them, a process known as shuttling. Furthermore, a modular architecture based on moving the ions around different regions of the trap was proposed, with the aim to enable scalability. This modular structure enables parallel gate operations and scalable quantum charge-coupled device (QCCD) architectures [65]. Shuttling the ions is a fundamental process that allows all-to-all connectivity in ion trapping devices. However, motional heating can occur during ion shuttling and this needs to be mitigated by using extra cooling steps. Therefore, both short

and long range transport, and laser cooling have significant impact in the duty cycle and therefore the power consumption.

The state of the qubits is measured by exciting the ions into an auxiliary excited state that rapidly emits light (a cycling transition), which can be captured by a camera or a photomultiplier tube [66]. Ion traps can be implemented at room temperature; the ions are isolated from the external noise using ultra high vacuum (UHV) and cooled down using laser cooling. However, ion trap chips [67, 68] are often operated at cryogenic temperatures in order to reach a better vacuum conditions and suppress electric noise from the trap.

Our baseline analysis considers a generic, linear, QCCD architecture, operating at room temperature. We refrain from choosing laser or microwave gates, and instead use an average gate drive which would encompass both. The time duration of the gates are based on laser gates, but are of the same order of magnitude of the fastest microwave gates. A single ionic species is assumed to be used, and no sympathetic cooling is performed, only Doppler and sideband cooling [69, 70].

5.2 Energy model

This section discusses the energy efficiency of a trapped-ion device constructed from the hardware elements displayed in Appendix A.3. As a baseline, we consider a trapped-ion computer that has a single gate zone, which means it can only perform one single- or two-qubit gate at a time. In the next section, the analysis is extended to architectures with multiple gate zones where parallelization is possible, in order to study the potential increase in energy efficiency they enable by decreasing computation time.

5.2.1 Power consumption of the hardware

The detailed list of components of a trapped-ion device with a single gate zone is shown in Table 6. The majority of data in this table has been measured in the NQCC trapped ions laboratory. The power consumption related to the Heating, Ventilation and Air-Conditioning (HVAC) system, which stabilizes the temperature and humidity required by the laser systems, corresponds to an average estimation based on both our trapped ions and neutral atoms data, assuming they require similar conditions. However, HVAC power consumption strongly depends on the facilities hosting the computer. In this work, a value was chosen in accordance with observations in real facilities, but this value might vary depending on the efficiency of the entire infrastructure, the size of the room, the climate conditions of its location, etc. Moreover, the following estimations do not include the cryogenic setup that is used in the NQCC laboratory, based on the fact that most atom-based devices do not operate at cryogenic temperatures.

Figure 9 shows the consumption of the hardware listed in Table 6 in three groups: Environmental Conditions, Qubit Control and Classical Processing. This particular device would be able to host the order of ~ 10 qubits, but we expect the same order of magnitude in power demand for a ~ 100 qubits computer.

According to our estimations, the overall consumption of this particular device is $P_{\text{trapped ions}} \approx 10$ kW, which corresponds to ≈ 900 MJ of energy spent in 24 hours. As one can see, there is a large gap between the power needed to ensure the suited environmental conditions and the rest of the components. This is similar to the high demand of cryogenic fridges we see in solid-state platforms. In particular, the consumption of the HVAC system represents 78% of the total consumption of the device. This value, however, can be highly variable for the reasons we comment above. In addition, this computer can be expected to function under a less demanding HVAC system, since its laser setup is lighter than the one included for the neutral-atom computer in the next section.

The aggregated consumption of all the qubit control components is ≈ 1.9 kW. Similarly to previous platforms, there is a clearer relation between the size and power consumption of the qubit control components and size of the quantum processor. In this case, the components are mainly determined by the number of gate zones in the device and the ability to perform independent gates on those gate zones. This is further discussed in Appendix A.3.

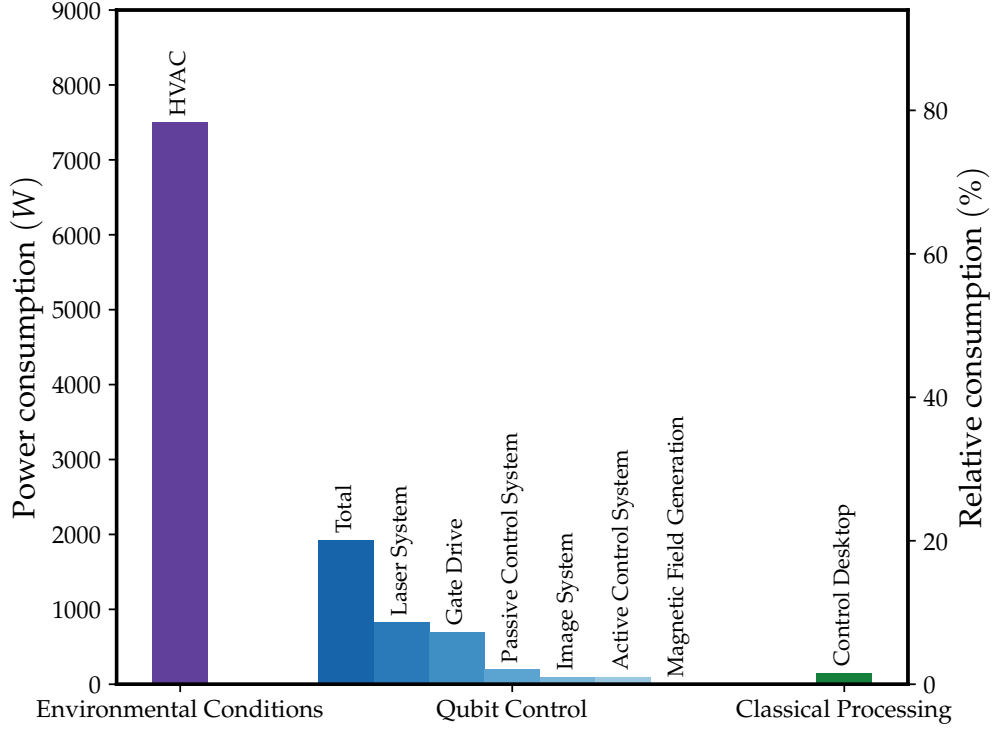


Figure 9: Power breakdown of a trapped ions computer, according to the components shown in Table 6. In the x-axis, the components are grouped depending of their purpose in the computer. Each group has a bar that corresponds to the total accumulated power of its components. The left y-axis shows the absolute value of power consumption of the components, and the right y-axis the relative value with respect to the overall consumption of the computer.

5.2.2 Derivation of the energy efficiency

This section explains the model used to estimate the running time of an algorithm $\mathcal{A}$ composed of N_{samples} shots of a quantum circuit with depth D . As explained in the introduction, the running time $t_{\mathcal{A}}^{\text{trapped}}$ of an algorithm is given by the running time $t_{\text{circuit}}^{\text{trapped}}$ of a quantum circuit applied on a trapped ion device:

$$t_{\mathcal{A}}^{\text{trapped}} = N_{\text{samples}} \cdot t_{\text{circuit}}^{\text{trapped}} = N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}} + \beta_{\text{trans}} \cdot D \cdot t_{\text{trans}}), \quad (16)$$

where N_{samples} is the number of shots, t_{reset} is the time necessary to reset the qubit between every shot of the algorithm, t_{meas} is the time needed to read the final state of the qubits at the end of the circuit and t_{clock} is the clock time of the computer. For the most part, the model is equivalent to the solid-state one, with the addition of the time t_{trans} needed to transport the atoms from the storage to the interaction zone. We introduce the parameter β_{trans} , the ratio of the number of layers requiring long-range transports over the total depth. This parameter characterizes how many times qubit need to be moved in and out of the gate zones and highly depends on the circuit, and it is further discussed in Appendix B.2.

In order to provide a state-of-the-art estimation of the computing time, the following values are based on Refs. [71, 72], which demonstrate experimentally the viability of QCCD computers and provide values of their operational times. However, we adjust their numbers in order to have a general representation of a trapped ions device that have a storage zone connected to a single, or multiple, gate zones distributed linearly. First, we consider $t_{\text{reset}} \approx 150$ ms, which corresponds to the time needed to move the qubits from the gate zones to the storage zone (50 ms) and the time to perform Doppler and sideband cooling

to reset the qubits into their $|0\rangle$ state (100 ms).

There are different shuttling operations that need to be performed to operate with the ions in the different gate zones. First, we consider the transportation time $t_{\text{trans}} \approx 50$ ms, which is the result of shuttling the ions from the storage zone and the gate zone, and any additional cooling necessary before performing the gates. Next, there is a shuttling operation needed to move the ions within the gate zone, if the device has multiple gate zones. This time is typically shorter than the transport time, and depends on the number of gate zones in the computer, as well as the velocity at which the ions can be moved and the distance between zones. For multi-zones devices, this operation dominates the clock time, and can be estimated by $t_{\text{clock}}^{\text{multi-zone}} \approx 250 \mu\text{s} \times (N_{\text{zones}}^{\text{gate}} - 1)$. When it comes to the single-gate-zone device, this shuttling is not needed, thus we consider its clock time to be the time needed to perform two-qubit gates: $t_{\text{clock}}^{\text{single-zone}} \approx 170 \mu\text{s}$. In this case, this value accounts for a two-qubit entangling gate. Finally, the state of the qubits is measured using a digital camera or a photomultiplier tube (PMT), that take $t_{\text{meas}} \approx 0.5$ ms.

In the case of a single-gate-zone device, t_{trans} is three orders of magnitude larger than t_{clock} . Therefore, the execution time of a circuit in this device depends highly on the ratio of transports β_{trans} . For instance, a sample of a circuit of depth $D = 100$ with $\beta_{\text{trans}} = 1$ would take ≈ 5.1 s. This circuit corresponds to a series of gates that are always applied on different qubits, *i.e.* the device has to bring a new qubit (or pair of qubits) from the storage to the gate zone every time step. In contrast, the time to execute a circuit with $\beta_{\text{trans}} = 0.5$, is ≈ 2.6 s. In this case, the circuit performs, on average, two consecutive gates to the same qubit (or pair of qubits) before reaching for a new one from the storage zone. The computing time of a circuit with no transport ($\beta_{\text{trans}} = 0$) would be ≈ 0.1 s. This circuit would be a series of 100 gates applied on the same qubit, but serves as an example of how the structure of the circuit can affect the final execution with such a limited number of gate zones.

Taking all of this into account, the equation for the energy efficiency of trapped-ion devices is :

$$EE^{\text{trapped}} = \frac{1}{N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}} + \beta_{\text{trans}} \cdot D \cdot t_{\text{trans}})} \cdot P^{\text{trapped}}, \quad (17)$$

where P^{trapped} is the sum of the power consumption of the hardware elements of Table 6.

5.3 Results and discussion

Figure 10 shows the energy efficiencies and number of computations in 24h of a single-gate-zone device, depending on the depth of the circuit and number of samples.

For the reasoning explained above, we assume that the circuits in Figure 10 have a ratio of transport-per-layer $\beta_{\text{trans}} = 1$, *i.e.* the device performs long-range transportation every layer. Effectively, this would correspond to $t_{\text{clock}} \approx 50$ ms. This causes the profile of the curves in panel (a) to be inversely proportional to D , since the time to execute a single time-step of the circuit is close to the reset and measurement time of the qubits. Therefore, even for short circuits, the transport time takes over the total computation time.

Now, we want to explore the potential energetic advantage of adding additional gate zones in order to parallelize the gates of the circuits. In principle, this would highly decrease the depth of the circuits, thus optimizing the energy efficiency of the device. In this case, we do not expect an increase of hardware components due to the addition of gate zones. In fact, it would only require a negligible increase of power supply to the laser system. However, in order to take full advantage of all the gate zones in the device, we include one more Active Control System, which enables the device to perform single- and two-qubit gates at the same time. Otherwise the device would be limited to perform a single gate type per layer, and the depth of the circuits would increase. In addition, we also estimate the clock time to be longer than the single-gate-zone device. As we explain above, $t_{\text{clock}}^{\text{multi-zone}} \approx 250 \mu\text{s} \times (N_{\text{zones}}^{\text{gate}} - 1)$. For the 5-zone device, the clock time would be $t_{\text{clock}}^{5\text{-zone}} \approx 1000 \mu\text{s}$, whereas for the 10-zone device it would be $t_{\text{clock}}^{5\text{-zone}} \approx 2250 \mu\text{s}$. Both of these times are still an order of magnitude below the transport time, and we can expect the transport to be the most time-consuming operation for circuits with β_{trans} close to 1.

In Figure 11 we display the energy efficiencies of devices with 1, 5 and 10 gate zones, executing one sample of circuits with pre-routing depth of $D_0 = 500$ layers. The results depend on the average of gates

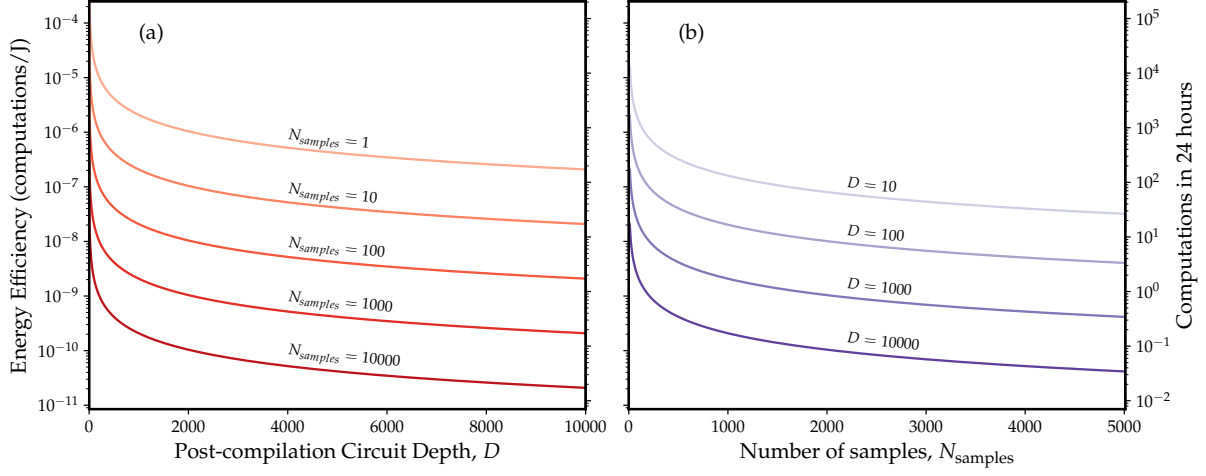


Figure 10: Energy efficiency (left y-axis) and number of computations (right y-axis) in 24 hours of a single-gate-zone trapped-ion computer depending on the final circuit depth, D and number of samples, N_{samples} . Panel (a) shows several series with different N_{samples} values and the x-axis corresponds to D . The series in panel (b) have different D values and its x-axis corresponds to N_{samples} . The circuits displayed in the panels have a ratio of transport per layer $\beta_{\text{trans}} = 1$.

per layer $\overline{N}_g^L$ and the ratio of transports β_{trans} . The upper panel of Figure 11 shows the minimum value of the ratio β_{trans} of the three devices, depending on $\overline{N}_g^L$. Here, we assume that, if the average number of gates per layer is higher than the number of gate zones, the device is forced to perform a certain amount of transports. For instance, in order to execute a circuit with $\overline{N}_g^L = 20$ gates per layer on the $N_{\text{zones}} = 10$ gate-zone device, the original layers of the circuit will have to be split in two. Since the 20 gates of the original layers use necessarily different qubits, the device will be forced to change the qubits between the two new layers that result from splitting the original ones. In Appendix B.2, we detail how we compute this minimum value of β_{trans} . On the contrary, the maximum value of this ratio can always be $\beta_{\text{max}} = 1$.

In the lower panel of Figure 11, we show the envelopes between the worst- and best-case scenarios of transport. The upper bounds of these envelopes correspond to the minimum value β_{min} , while the lower bounds correspond to $\beta_{\text{trans}} = 1$, *i.e.* all layers of the circuit require transport. There is an extreme case where the single-gate-zone device would outperform the others, when $\overline{N}_g^L = 1$ and $\beta_{\text{trans}} = 0$. This corresponds to a circuit that only performs gates on a single qubit (or pair of qubits). In this particular case, the assumption that we make above, where we consider the clock time of the multiple-gate-zone computers to be equal to the transport between gate zones does not hold, since this operation is not needed. To make the comparison between devices fair, Figure 11 also includes the energy efficiencies of the multi-gate-zone devices with $t_{\text{clock}} \approx 170 \mu\text{s}$, where its improvement is highlighted with a black pattern. Then, the energy efficiency of the single-gate-zone computer is negligibly better, due to the 100 W saved with respect to its counterparts. That being said, there are virtually no algorithms that can be executed faster in this device; even for extremely well-suited circuits, a device with multiple gate zones can always perform the same operations at least with the same speed.

As the density of the circuit grows, the non-existing parallelization capacity of the single-gate-zone device results in a necessary depth overhead. Moreover, the device has to constantly move qubits from the storage to the single gate zone, which raises the amount of transports that it has to perform. These circumstances make the efficiency of the single-gate-zone device to drop drastically for $\overline{N}_g^L > 1$ gates per layer.

When it comes to the computers with 5 and 10 gate zones, one can see how they are slightly apart for circuits with $\overline{N}_g^L < 5$ and $\beta_{\text{trans}} = 0$, due to the fastest clock time of the 5-zone computer. However, since this clock time corresponds to the time to move the ions between gate zones, we assume that, with a good compilation strategy, the same clock times could be achieved for the 10-zone device, by restricting

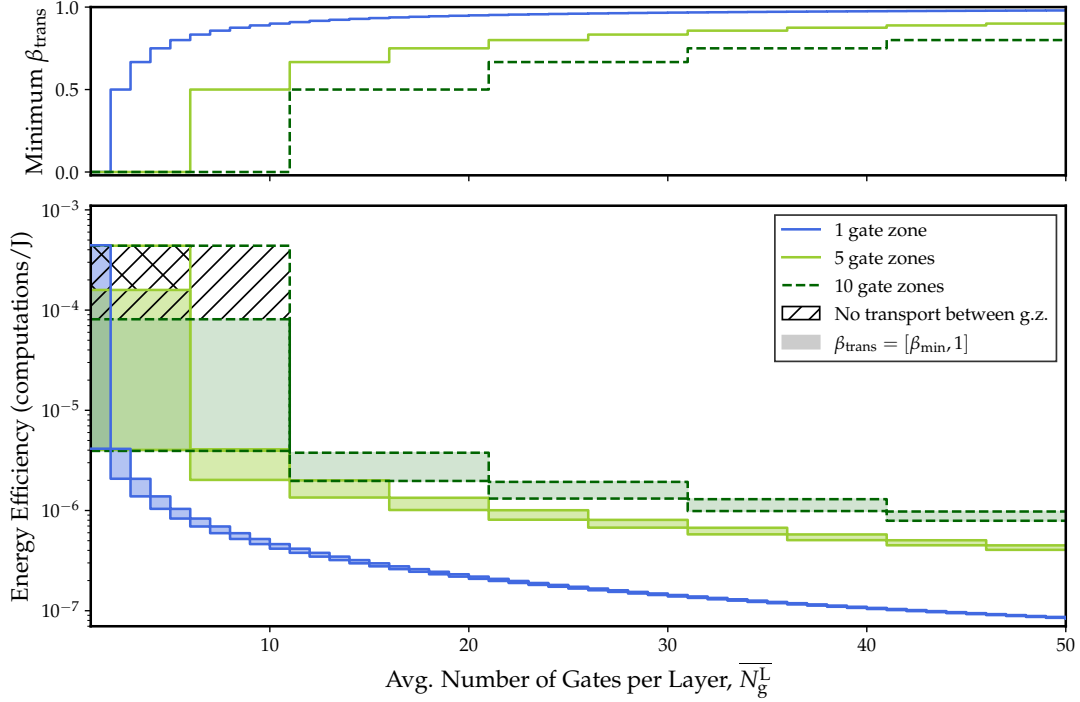


Figure 11: The upper panel shows the minimum value of the transport per layer ratio β_{trans} for the 1, 5 and 10 gate-zone devices, according to the average number of gates per layer $\overline{N_g^L}$. The lower panel shows the energy efficiencies of such devices when running a single sample of a $D_0 = 500$ circuit. The shaded areas display the possible efficiencies between the worst- and best-case scenarios $\beta_{\text{trans}} = [\beta_{\text{min}}, 1]$. The black crossed area represents the improvement of the multi-gate-zone devices for circuits that transport between gate zones, for which $t_{\text{clock}}^{\text{multi-zone}} = t_{\text{clock}}^{\text{single-zone}} = 170 \mu\text{s}$.

the amount of gate zones used to 5, similarly to what we discuss above for $\overline{N_g^L} = 1$. The efficiencies of these devices do not drop until $\overline{N_g^L} > N_{\text{zones}}$. Before that, there is no overhead due to their limited number of gate zones, assuming an even distribution of gates per layer through the circuit. Although there is an overlapping area of efficiencies between the two devices, a given circuit might have different values of β_{trans} for each of them. For instance, the well-known GHZ state preparation circuit [73], which has a staircase structure with one CNOT gate per layer, will require two times more transport for the 5-gate-zone device than the 10-gate-zone one.

As the average number of gates per layer increases, the minimum value of β_{trans} grows, and the area between the worst- and best-case scenarios shrinks. Interestingly, the efficiency gap between the multi-zone devices is never as large as the gap between β_{min} and $\beta_{\text{trans}} = 1$ whenever $\overline{N_g^L} < N_{\text{zones}}$. This result leads to the conclusion that a relevant advantage is induced by devices with large number of gate zones, enough to execute the majority of layers of the circuit without adding extra layers and transport. Moreover, the use of advanced compilation techniques is key to optimize the ratio of transport β_{trans} and reduce the execution time of the circuit. Since the power demands of an HPC-based compiler could lead to a significant increase in consumption, there is a trade-off to consider between the energy spent in classical optimization and the one used by the quantum device itself.

6 Neutral atoms

Neutral-atom quantum processors, much like trapped-ion systems, encode quantum information in individual atoms. Unlike ions, however, neutral atoms carry no net charge and therefore are not trapped

via electric fields. Ordered lattices and tweezer arrays of neutral atoms have developed into a leading platform for large scale analog simulation, where they provide highly controllable realizations of many-body systems [74]. More recently, advances have been made to perform digital quantum computation with neutral atoms trapped in optical tweezers via discrete Rydberg entangling gates [75, 76, 77]. These experiments have emerged as a promising platform for quantum computation and have demonstrated error-correction [78, 79] and digital simulation over hundreds of qubits [80, 81]. Commercial adoption of neutral atoms has become increasingly widespread with the establishment of companies such as Pasqal, QuEra, Atom Computing, Infleqion and many others.

6.1 Technology

In neutral atom devices, individual atoms, most commonly Alkali [78] and Alkali-earth atoms [82, 83, 84], are trapped and laser cooled in the tweezer potentials of tightly focused beams of light. Each choice of atomic species, provides energy levels that can be used to store and manipulate quantum information. Although a variety of approaches exists and most of the following discussion is generally applicable, we will specifically consider the example of using rubidium-87 atoms for digital quantum computation.

Tweezers are initially loaded from a cloud of laser-cooled atoms in a magneto-optical trap (MOT), prepared inside and ultra-high vacuum chamber. In digital neutral atom processors, qubits are encoded within the ground state hyperfine manifold of the atom. These states offer long coherence times ($T_2 > 2s$) during which quantum circuits can be executed. Quantum gates are implemented by irradiating atoms with specific laser pulses. Arbitrary single-qubit rotations are performed within the hyperfine manifold using qubit-specific, parallel Raman excitations. Entangling operations are performed by exciting atoms into highly-excited atomic Rydberg states. Then, the entangled qubits can be controlled by either only illuminating certain atoms within a dense 2D lattice with local Rydberg beams [81], or by rearranging atoms with movable beams between each gate layer [85]. Only atoms placed within the Rydberg blockade radius will be entangled. In the latter mode, atoms are trapped in two types of tweezers: static tweezers generated via phase manipulation with a spatial light modulator (SLM) or movable tweezers manipulated on the time-scale of circuit operations in a highly parallel way via acoustic-optic deflectors (AODs). After the computation, qubit states are readout on a camera via local or global fluorescence imaging through an objective.

In summary, qubits are encoded in the energy levels of laser-cooled neutral atoms. Universal quantum computation is achieved by performing single and multi-qubit gates by exciting atoms via specific laser pulses. Furthermore, by shuttling atoms within the computational region, a highly-programmable connectivity between qubits is accessible, enabling crucial tools of error-correction and algorithms such as non-local computation and transversal gates.

6.2 Energy model

This section discusses the hardware elements of a neutral atom quantum computer and the expected power consumption of each component. To ensure comparability with other platforms, the following table has been estimated for a system of $N_q = 400$ qubits, although several experiments have demonstrated using several thousands of qubits [86, 87]. We also specifically consider the power requirements for a neutral atom processor based on rubidium-87 with entangling operations performed through a two-photon Rydberg transition where atoms are rearranged between gates, although these considerations should be broadly applicable.

6.2.1 Power consumption of the hardware

Current neutral atom quantum computers setups are dominated by free-space optics with often considerable laser power requirements. These laser systems are either entirely free-space or delivered to the atoms via fibers. Its important to note that as error-rates improve, it becomes increasingly clear that maintaining the stability of these various macroscopic optical components is crucial to achieving algorithmically relevant error rates [88] In addition to laser requirements and vacuum components, we therefore

also highlight the HVAC requirements to ensure stability in room temperature and humidity, necessary for maintaining alignment of many precise beams paths. As it is discussed in the trapped ions section, the power consumption of the HVAC system might vary heavily depending on the facilities hosting the quantum computer. In this work, we chose a mean value between the data measured for the NQCC trapped-ion device and the Harvard neutral-atom device. The HVAC system will likely remain a key, and potentially increasing, form of power consumption until technological stability advances, such as on-chip photonic circuits, are more readily available at the required optical powers.

The overall power consumption of the considered device is broken down in Figure 12. This figure shows the power consumption of all the hardware components displayed in Table 8 in three different groups: Environmental Conditions, Qubit Control and Classical Processing.

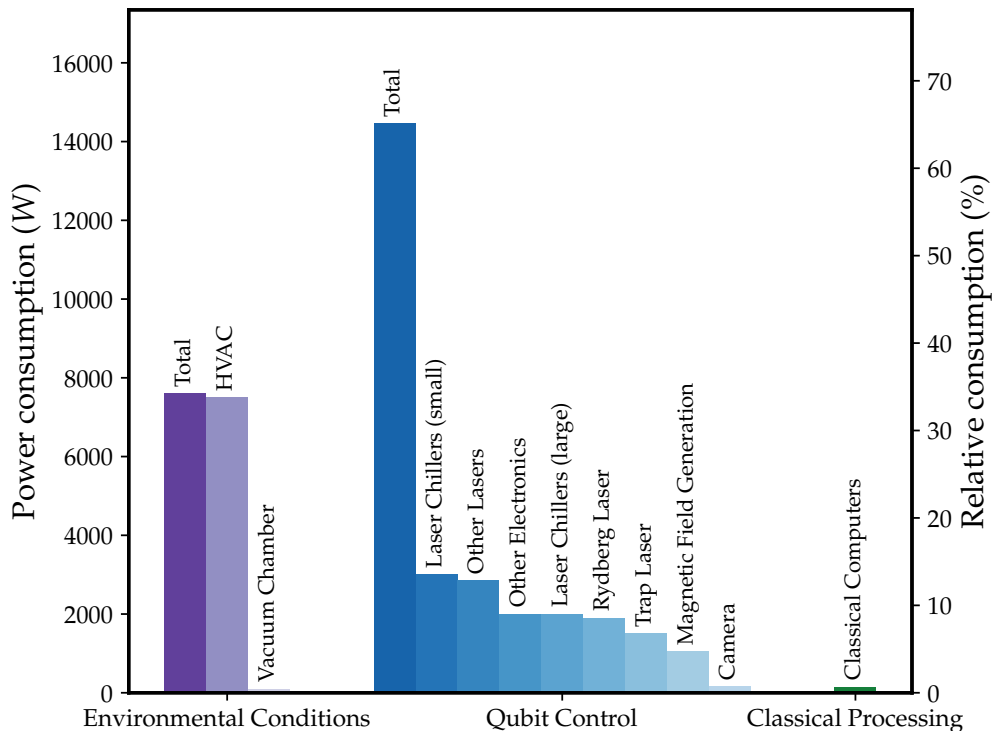


Figure 12: Power breakdown of a neutral-atoms computer, according to the components shown in Table 8. In the x-axis, the components are grouped depending of their purpose in the computer. Each group has a bar that corresponds to the total accumulated power of its components. The left y-axis shows the absolute value of power consumption of the components, and the right y-axis the relative value with respect to the overall consumption of the computer.

With these components in mind, the overall power consumption of the computer is $P^{\text{neutral}} \approx 22$ kW, which corresponds to 1900 MJ of energy consumed in 24 hours. In contrast with all the other platforms included in this work, the energy expenditure of this neutral-atoms computer is not majorly devoted to produce the suited environmental conditions for the qubits. In particular, components consume up to 7.6 kW of power, which corresponds to the 35 % of the overall consumption. In contrast with solid-state and photonic computers, neutral-atoms computers do not require a cryogenic setup to keep the chip enclosed in cryogenic temperatures. However, atoms that carry quantum information are still effectively "cooled" using lasers present in the Qubit Control Group.

In contrast, the power consumption of the Qubit Control group is 14.5 kW, which corresponds to 65 % of the overall consumption. In this group, we include different chillers, that evacuate the heat dissipated by the lasers, but are not devoted to cool down the register itself.

6.2.2 Derivation of the energy efficiency

In order to estimate the energy efficiency of a device based on neutral atoms, we now explain our model to estimate the time it takes to perform an algorithm $\mathcal{A}$ composed of N_{samples} of a quantum circuit with depth D . Although the technology is similar to the one of trapped-ion circuits, neutral atoms architectures impose different constraints on the running time of algorithm. In particular, the device needs to periodically refill atoms that are typically lost during the computation process. It has been shown that it is possible to load several thousands of atoms with a single set of tweezers directly from the MOT [86]. There, qubits can be addressed and gates can be performed. However after some time, new qubits need to be reloaded into the optical traps, adding a time t_{reload} [80, 89]. It was observed that, on average, 40 to 80 samples of a circuit can be run before requiring to refill the traps. Moreover, the current state-of-the-art allows to run circuits of depth $D \sim 30$. As a broad estimate, we set to $30 \times 80 = 2400$ the number of layers after which a time t_{reload} is added to the circuit running time, which reads:

$$t_A^{\text{neutral}} = N_{\text{samples}} \cdot t_{\text{circuit}}^{\text{neutral}} = N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}} + \beta_{\text{trans}} \cdot D \cdot t_{\text{trans}} + t_{\text{reload}} \cdot \frac{D}{2400}), \quad (18)$$

where t_{reset} is the time necessary to reset the qubit between every shot of the algorithm, t_{meas} is the time needed to read the final state of the qubits at the end of the circuit, t_{clock} is the clock time of the computer, β_{trans} the ratio of layers of the circuit requiring moving the atoms to and from interaction zones and t_{trans} the transport time. For more details on β_{trans} , we refer the reader to the Appendix B.2. Equation 18 assigns the corresponding fraction of the reloading time that takes place every 2400 layers to the individual samples of the circuit. For instance, during the execution of circuits with 1600 layers, the register would be reloaded every 1.5 circuit samples. Therefore, each circuit execution would carry $1600/2400 = 2/3$ of the total reloading time. Here, we assume that one can reload the register during a circuit execution. This would require the quantum information to be copied on the new set of atoms (e.g. using SWAP gates), which we assume to take a negligible amount of time compared to the 500 ms spent on the reloading itself. In Ref. [90], the authors describe multiple experiments being run on a neutral atom device, with different operation times. For our estimations, we consider $t_{\text{reset}} \approx 10$ ms and $t_{\text{meas}} \approx 10$ ms. We assume that most of the time taken during every layer is devoted to configure the atom array inside the interaction zone according to the gates to be performed, which leaves us with $t_{\text{clock}} \approx 500$ μ s. Finally, we consider the time needed to transport atoms from the storage to the interaction zone is $t_{\text{trans}} \approx 500$ μ s.

Taking into account that the reload process happens every 2400 layers, the reloading time per layer of the considered computer is $t_{\text{reload}}/2400 \approx 210$ μ s. Therefore, the operations that depend on the depth of the circuits are two orders of magnitude below the measurement and the reset times. With this in mind, one can expect the latter to dominate the circuit time until $D \sim 100$. From that point, the other operations start taking over the computing time, until t_{circuit} becomes proportional to the depth. To put this into perspective, a single sample of a $D = 10$ layers circuit would take $t_{\text{circuit}} = 27$ ms, mostly dominated by $t_{\text{reset}} + t_{\text{meas}} = 20$ ms. For larger circuits with $D = 100$ and 1000 layers, the circuit time grows significantly to $t_{\text{circuit}} = 91$ and 720 ms, respectively.

The energy efficiency of a neutral atom quantum computer can be written as:

$$EE_A^{\text{neutral}} = \frac{1}{N_{\text{samples}} \cdot (t_{\text{reset}} + D \cdot t_{\text{clock}} + t_{\text{meas}} + \beta_{\text{trans}} \cdot D \cdot t_{\text{trans}} + t_{\text{reload}} \cdot \frac{D}{2400}) \cdot P^{\text{neutral}}}, \quad (19)$$

where P^{neutral} is the sum of the power consumption of the elements in Table 8.

6.3 Results and discussion

Figure 13 shows the energy efficiencies and number of computations in 24h of a the neutral atom device described above, depending on the depth of the circuit and number of samples.

In accordance with Eq. 19, the left panel of Figure 13 displays equidistant lines for the different samples values, indicating that EE is inversely proportional to N_{samples} . On the contrary, we see the $D = 10$ and 100 lines are closer to each other, with respect to the rest of them, in the right panel of the figure. As it discussed above, for small circuits, the reset and measurement times are dominant in the

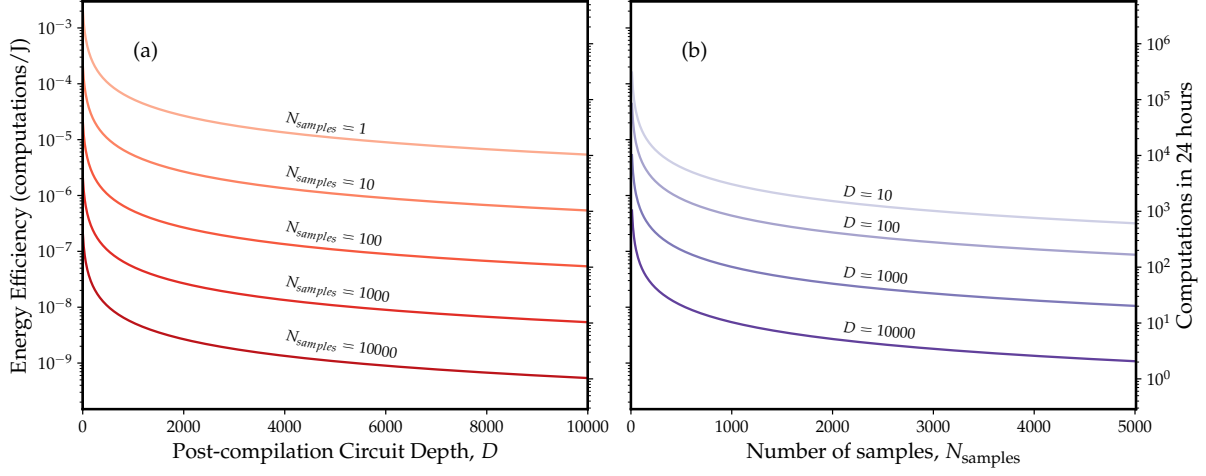


Figure 13: Energy efficiency (left y-axis) and number of computations (right y-axis) in 24 hours of a neutral-atoms computer depending on the post-compilation circuit depth, D and number of samples, N_{samples} . Panel (a) shows several series with different N_{samples} values and the x-axis corresponds to D . The series in panel (b) have different D values and its x-axis corresponds to N_{samples} . The circuits displayed in the panels have a ratio of transport per layer $\beta_{\text{trans}} = 0.25$ transports per layer.

overall circuit time. This makes circuits around $D \sim 10$ to take similar t_{circuit} . As the depth grows past 100, the EE of the neutral-atom computer becomes inversely proportional to D . Note that a value of $\beta_{\text{trans}} = 0.25$ transports per layer have been chosen, corresponding to circuits where atoms are moved to the interaction zone every four layers on average. This value can be optimized and is highly dependent on the algorithm, but its effect is more trivial than in the case of trapped ions due to the lower transport time in neutral atom devices.

To scale neutral atom system to large, practical algorithms it has become increasingly clear that the atom array must be loaded and used continuously. In a few recent experiments this has been executed by laser cooling atoms in one chamber and then transporting them via, for instance, an optical lattice to be loaded into an array of prepared qubits [87, 91, 92]. This modality allows for fast circuit cycle rates and can be scaled to longer computations. We therefore consider it as an extension of the energy model. Essentially, the terms related to the reloading and resetting of atoms in Eq. 18 vanish and, in return, additional components, for instance an additional transport lattice and more complex vacuum systems, are considered in the total power consumption of the device. These additional hardware components, shown in Table 8 of Appendix A.4, increase the power consumption of the studied device by 2550 W.

Figure 14 shows the energy efficiency comparison between a neutral-atoms computer that loads its register periodically and a similar computer that has the ability to load the register continuously.

As one can see, the computer that reloads the register continuously displays higher energy efficiencies for all D values, regardless of its extra power consumption. For short circuits, the reset and measurement operations dominate computing times. In that regime, the continuous reload computer needs 10 ms less time for each sample, since it is always refilling the register with atoms prepared at state $|0\rangle$. However, the higher speed of this computer overcompensates the 2550 W power increase with respect to its counterpart. As the depth of the circuits grow, the energy efficiency gap increases, until it surpasses the $D \sim 500$ layers. This regime corresponds to the moment where EE starts becoming inversely proportional to the D . From that point, the improvement ratio between the computers tends asymptotically to $EE^{\text{cont}}/EE^{\text{periodic}} \rightarrow 4/3$ (for $\beta_{\text{trans}} = 0.25$). The ratio of improvement $EE^{\text{cont}}/EE^{\text{periodic}}$ does not depend on the number of samples executed.

Figure 14 includes circuits with depths past the reloading frequency $D = 2400$, which, in order to be executed in the Periodic Reload computer, would need to perform the reloading operation in the middle of a sample. As mentioned above, we assume that this could be done by swapping the information

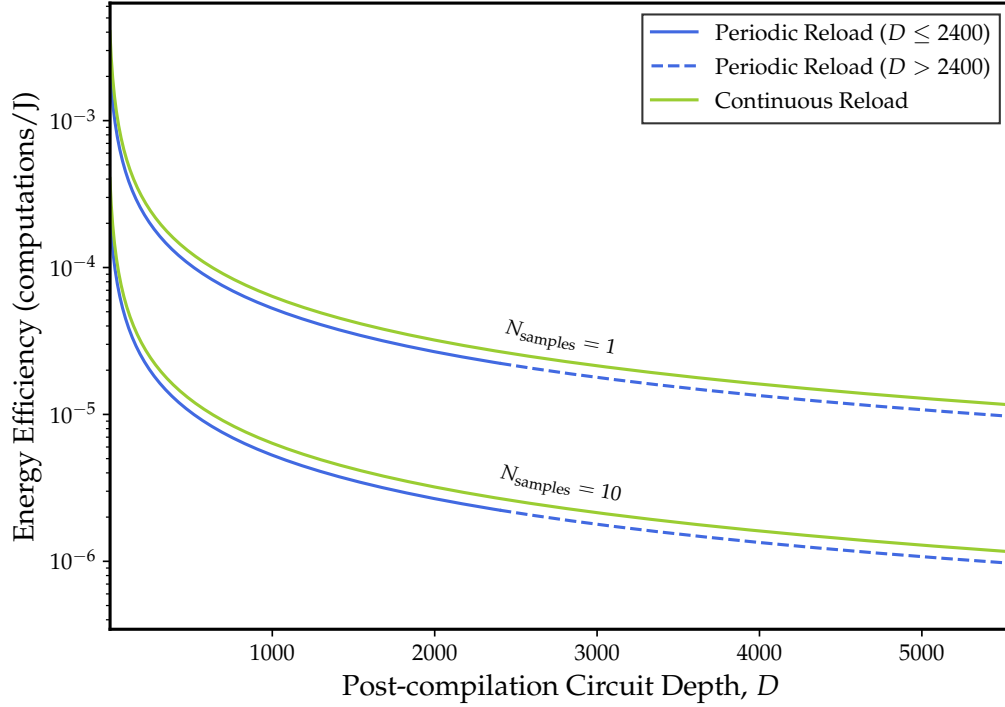


Figure 14: Energy efficiency of a neutral-atoms computer executing 1 sample of a circuit, depending on its post-compilation depth. The blue curve corresponds to a computer that reloads its atom periodically, and the green line to a device that can reload them continuously, without interruption. The dashed blue line corresponds to those circuits that surpass the reloading frequency $D = 2400$, and thus would need to perform the reloading operation during its execution. All the circuits displayed have a ratio of $\beta_{\text{trans}} = 0.25$ transports per layer.

to the newly set of atoms entering the register, but this process might impose new physical challenges and constrains to these computations, which would probably affect the computing time and fidelity of the algorithm. Our results show an efficiency improvement due to the adoption of the continuous reload technique in neutral-atoms computers. This could be even more significant under a fault-tolerant computation scheme, where qubits are measured and reset after each syndrome detection round, creating windows for constantly substituting the used atoms with freshly loaded ones.

7 Photonic computing

Photonic quantum computing architectures represent a paradigm change compared to the other platforms studied in this article. Unlike matter-based platforms, photonic quantum computing uses light to perform computation. This approach has the advantage of avoiding the relatively short coherence time of matter-based qubit compared to photonic ones. In addition, the manipulation of light using linear optics components can be done at room temperature, which reduces the cost of cryogenic power. However, these relative advantages are counterbalanced by the difficulty of storing light for long periods of time.

There are two approaches to photonic quantum computing: continuous variable (CV) and discrete variable (DV). In the former, information is encoded in the properties of the electromagnetic field, that can take any value in a continuous interval (e.g. position and momentum). In the latter, information is encoded in discrete modal properties, such as polarization or the spatial mode of single photons. For both schemes, operations require using of linear optics elements, such as beam splitters or phase shifters, but in order to perform universal computing, some additional elements are necessary. The CV scheme

requires non-Gaussian states of light, while in DV computing, a key ingredient is the generation of entangled single-photon states. This entanglement is introduced either through entangled cluster states, also called resource states, on which measurements are performed following the so-called measurement-based quantum computing paradigm [93], or through entangling gates using projective measurements [94]. The implementation of these schemes on physical hardware is currently being developed by several companies, such as Xanadu, implementing CV computing with Gaussian states of light [95, 96], PsiQuantum, implementing a novel, fault-tolerant, measurement based, DV computing scheme called fusion-based quantum computing [97], or Quandela, implementing linear optics DV quantum computing [98].

In this work, we focus on DV photonic quantum computing, and its implementation on quantum computers using quantum dot based single photon sources. For an overview of the field of photonic quantum computing, we refer to [99].

7.1 Technology

In DV photonic quantum computers, information is encoded in discrete degrees of freedom of single-photon states such as their path. The first essential building blocks of these computers are single-photon sources. These sources need to meet high efficiency requirements and deliver photons of near-unity purity and indistinguishability. The development of such sources is an active field of research and two main schemes are under development. On the one hand, probabilistic sources based on nonlinear materials, exploiting either spontaneous parametric down-conversion [100, 101, 102] or four wave mixing [103, 104] are being studied. In order to overcome the probabilistic nature of these mechanisms, heralding and multiplexing strategies need to be implemented which results in a high overhead in the resources required. On the other hand, deterministic sources using semiconductor based quantum dots can produce high quality photons on demand [105]. In this work, we focus on the latter kind of sources which are used in Quandela’s architecture [98].

In this setup, the source is operated at a temperature around 5 K and emits a temporal train of photons. The photons are then sent through an active demultiplexer followed by fibered delay lines which convert the train of photons into photons arriving simultaneously on an m mode photonic chip. The chip contains a universal interferometer, capable of implementing any $m \times m$ unitary matrix. Such an interferometer can be realized according to three different schemes which implement all-to-all connections between the modes using linear optics components : beamsplitters couple the spatial modes and phase shifters adjust their interferences. The first, due to Reck et al. is comprised of a triangular mesh [106]. The second, due to Clements et al. is comprised of a square mesh and, by a consequence of its symmetry, offers improved spatial footprint, stability and balanced losses [107]. The third, due to Pérez et al. utilizes a Hexagonal mesh which, while using more active elements, offers additional functionalities due to its ability to create recirculating structures [108]. Tunable beamsplitters are generally implemented via Mach-Zehnder interferometers with a phase shifter in one or both arms. The chip is then completely controlled by adjusting phase shifters and allows to implement any unitary matrix by following algorithms defined for each type of mesh. Note that these unitaries act on the optical modes, not on the photons directly. Finally, photons are detected using superconducting nanowire single photon detectors (SNSPDs), whose coincidence counts are connected to a correlator and a classical software. Depending on these outputs, a new electronic control signal can be applied to change the unitary applied to the next train of photons.

The time required to configure the chip depends on various parameters, for instance on the phase shifter technology. Thermo-optical phase shifters, for instance, require about 1 μ s while electro-optical phase shifters are much faster, around 10 ps (see Appendix A.5). In practice, configuration times are longer due to crosstalk between phase shifters.

7.2 Energy model

7.2.1 Power consumption of the hardware

In Table 10 of Appendix A.5 one can see the hardware components of a photonic quantum computer. In the architecture that we consider as a baseline, the number of most components does not increase

straightforwardly with the number of optical modes. For example, the number of sources needed to reach a large number of optical modes is related to the size of the demultiplexers. We consider here quantum dot sources and a demultiplexer able to send at most 12 photons on the chip per source, in line with current state of the art devices. Several sources can be operated simultaneously using a single laser using beam splitters. Note that the performance of the source, usually characterized by its brightness, and by the purity and indistinguishability of the photons produced, is independent of how much energy is used to operate it. Specifically, making a high quality source requires leveraging complex physical effects in cavity quantum electrodynamics and condensed matter, which represents a technological challenge when constructing the source, but operating the source always consumes the same amount of energy.

A demultiplexer consists of several delay lines which allow to synchronize the photons emitted sequentially by the source, so that they enter the chip simultaneously. For an m mode chip, a demultiplexer uses $m - 1$ Pockels cells, each consuming 0.5 W. Controlling these cells requires using an amplifier which uses about 4 W.

Superconducting nanowire single photon detectors (SNSPDs), on the other hand, scale with the number of modes. Although both sources and detectors operate at cryogenic temperatures, they can be placed in the same cryostat. We take as an example current small cryostats that can host up to one quantum dot source and ~ 25 SNSPDs per cryostat. For larger systems, larger cryostats would be more appropriate. We note that bigger cryostats have a larger power consumption, but are also more efficient: small air-cooled cryocoolers such as those considered here typically have an efficiency of 1% Carnot efficiency, while larger cryostats can reach 30-50% Carnot efficiency. In any case, if the number of modes m increases above the maximum number of SNSPDs that a cryostat can hold, we will need either more than one cryostat or a larger cryostat. Consequently, the cryogenic power does not increase linearly with the system size.

The chip on which the computation is performed requires strict control of the phase shifters and other active elements that apply. For our example architecture, controlling the chip efficiently requires monitoring its temperature and keeping it stable in order to avoid cross talk effects. Cooling the chip at room temperature requires a Peltier thermo-cooler; the power spent is related to the stabilization time of the phase shifters: for m modes, one needs $O(m^2)$ optical components. The control systems for the phase shifters are generally provided by field programmable gate arrays (FPGAs) due to their high speed and compact form factor. However, commercially available high-speed FPGAs are limited to 16 channels. As the number of active components grows so does the required number of FPGAs which require ~ 15 W each and can impose a significant power cost. The exact power cost of an FPGA is highly specific to the application but we can conservatively estimate the scaling of this power contribution as the number of electrical channels required to control N_q photonic qubits divided by the number of channels in one FPGA. This results in a factor of $\frac{(2N_q)^2}{16}$. Details of these classical control electronics relevant to the energy efficiency calculations may be found in Table 10.

7.2.2 Derivation of the energy efficiency

The framework that we have used in the previous sections to estimate the running time of algorithms does not hold for photonic computing architectures, since the computation happens very fast once the photon is created. After compiling an algorithm $\mathcal{A}$ into a circuit, the running time is still given by

$$t_{\mathcal{A}}^{\text{photonic}} = N_{\text{samples}} \cdot t_{\text{circuit}}^{\text{photonic}}. \quad (20)$$

However, a photonic circuit is not decomposed in a series of gates but rather seen as a single unitary that can be implemented in optical components arranged on a chip. The circuit running time is given by the time it takes for photons to go through the chip and be detected. Changing the depth of a circuit does change the length of this path, but the effect on the circuit time is negligible. However, as the depth grows, the probability that a photon is not lost while traveling through the chip decreases exponentially with the number of optical components. More precisely, the end-to-end transmission probability from the source to the detector is given by [109]:

$$\eta(m) = \eta_{\text{of}} \eta_{\text{d}} \eta_{\text{dmx}} 10^{-2c_{\text{coup}}/10} 10^{-D_{\text{optical}}(m)c_{\text{MZI}}/10}, \quad (21)$$

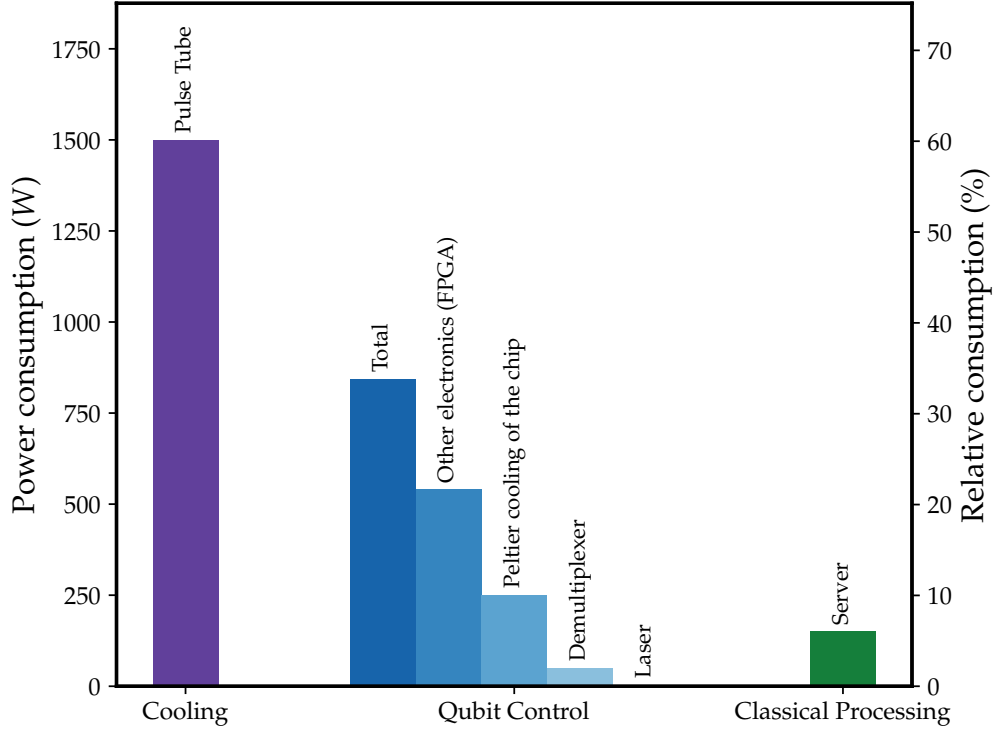


Figure 15: Power breakdown of a photonic quantum computer of 12 qubits (24 modes) according to the components shown in Table 10. In the x-axis, the components are grouped depending of their purpose in the computer. Each group has a bar that corresponds to the total accumulated power of its components. The left y-axis shows the absolute value of power consumption of the components, and the right y-axis the relative value with respect to the overall consumption of the computer.

where η_{of} is the efficiency of the source at the fiber output, η_d the efficiency of the SNSPDs, η_{dmx} the efficiency of the demultiplexer, c_{coup} the attenuation at the fiber to waveguide interface (there are two interfaces, hence the factor 2), c_{MZI} the attenuation per optical component, and $D_{\text{optical}}(m)$ the depth of the chip as a function of the number of modes m . The connection between (21) and photon loss is the following: given N input photons in a circuit of m modes, the transmitted photons follow a binomial distribution, of average $\eta(m)N$ and standard deviation $\sqrt{N\eta(m)[1-\eta(m)]}$. Moreover, the time necessary to detect N photons is proportional to $\eta(m)^{-N}$ (see (22) below).

In this work, we assume that the chip represents a universal interferometer, designed following the Clements architecture [107]. In this case, $D_{\text{optical}}(m) = m$, and the total number of components scales like m^2 . The values for the other components can be found in Table 11 and 12.

The average time necessary to successfully create one sample of a circuit on N_q qubits is given by

$$t_{\text{circuit}}^{\text{photonic}} = \frac{N_q}{N_{\text{source}} \cdot r_{\text{source}}} \cdot \frac{1}{\eta(2N_q)^{N_q}}, \quad (22)$$

where N_{source} is the number of single-photon sources and r_{source} their rates. There, two modes of light are used to encode a single qubit. The rate of the sources can be limited by either the lasers themselves or the detectors dead time, but not by the optical depth. Here, we choose the state-of-the-art value $r_{\text{source}} = 1$ GHz and assume that $N_{\text{source}} = 1$. The energy efficiency of photonic quantum computers of N_q qubits is thus given by

$$EE^{\text{photonic}} = \frac{1}{N_{\text{samples}} \cdot \frac{N_q}{N_{\text{source}} \cdot r_{\text{source}}} \cdot \frac{1}{\eta(2N_q)^{N_q}} \cdot P^{\text{photonic}}}, \quad (23)$$

where P^{photonic} is the total power consumption of the hardware presented in Table 10.

7.3 Results and discussion

In Fig. 16, we show the energy efficiency of photonic quantum computers as a function of the number of samples N_{samples} , for different number of qubits.

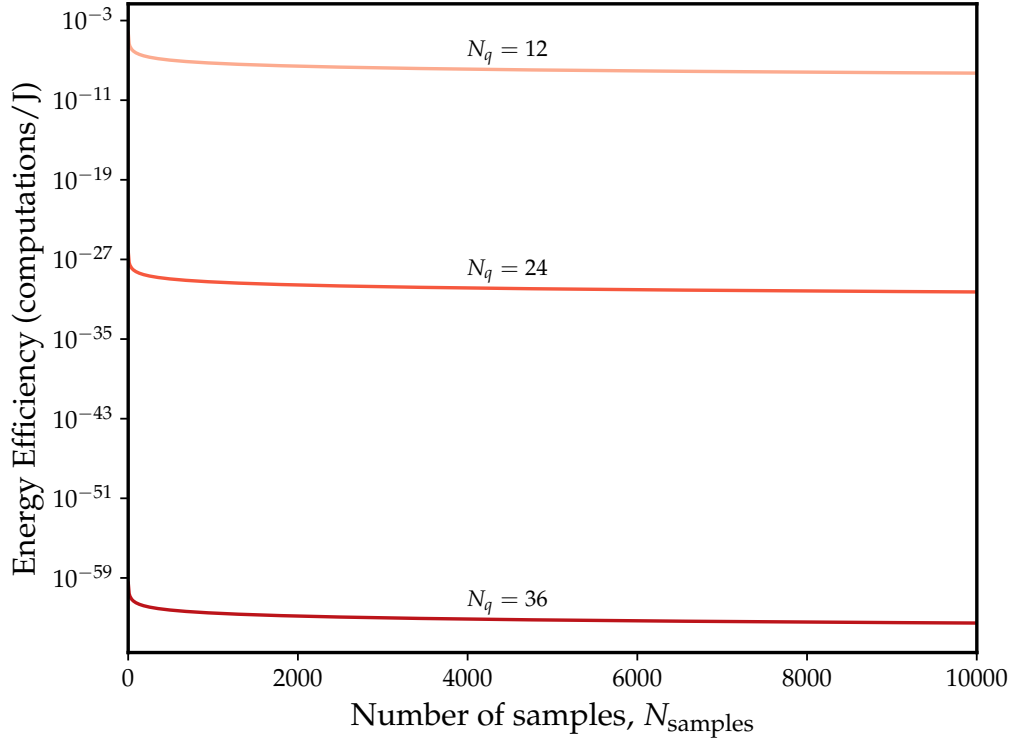


Figure 16: Energy efficiency of a photonic quantum computer as a function of the number of samples, for different number of qubits N_q .

From this plot, it can be immediately seen that current photonic architectures do not scale well with the number of qubits. This is mainly due to the quadratic scaling of the number of optical components on the chip as the number of modes grows. Not only does this lower the end-to-end transmission probability of the chip, but it also requires commensurate scaling of electronic qubit control components (mainly FPGA) to the list of hardware components. On top of these components, appropriate numbers of cryostats and demultiplexers are also added to match the required number of detectors. However, the current Quandela architecture with 12 qubits is very competitive in terms of energy efficiency.

Although today's photonic quantum computers are energy-dominated by cryogenics, the programmable interferometer mesh that implements the unitaries can also draw substantial power. As systems scale, this cost grows with the number of programmable elements, and different material platforms result in very different energy consumption. In Figure 17 we compare the state of the art in thermo-optic (TO) and electro-optic (EO) platforms and project the energy efficiency of each for a Clements interferometer mesh.

The optimal choice of phase shifter technology for photonic QC depends on regimes determined by the ratio of sampling and reconfiguration time as these correspond to dominance of static or dynamic components of power. The caveat is that the losses of the platform must not increase sampling time so much that additional cryogenic energy outweighs mesh savings. This is the regime of current technologies depicted in Figure 17, where the higher losses of EO platforms compared to TO platforms result in a

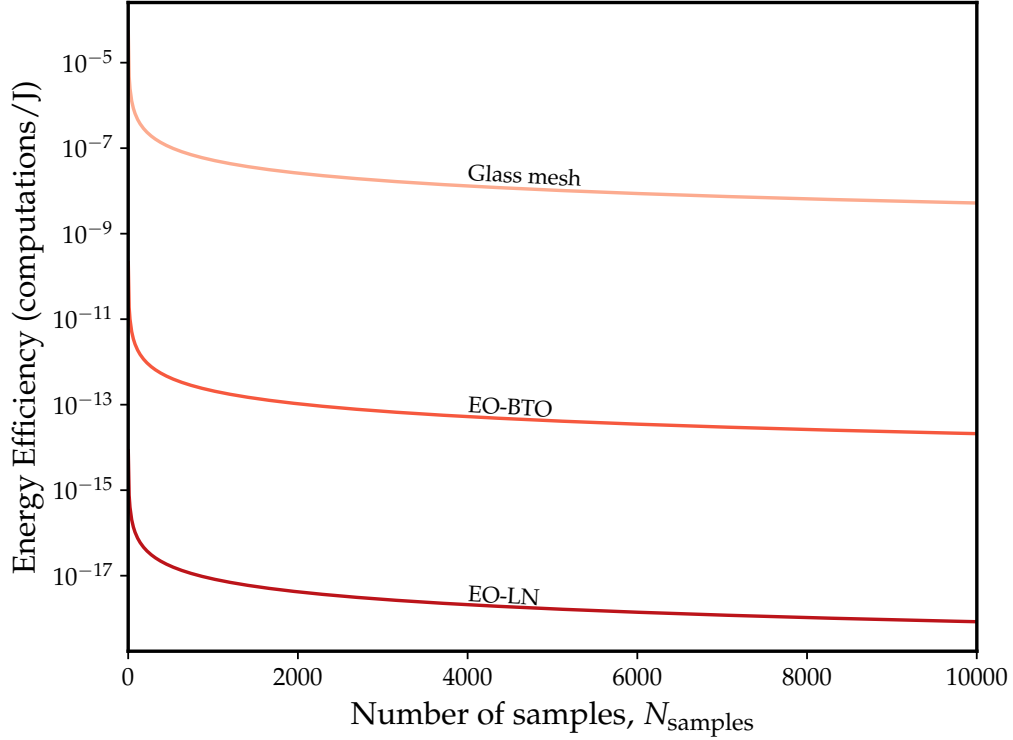


Figure 17: Comparison between state-of-the-art chip technologies assuming a 12-qubit photonic quantum computer in a Clements topology. Glass mesh refers to amorphous Silicon waveguides utilizing TO phase shifters, EO-BTO refers to Silicon Nitride waveguides evanescently coupled to Barium Titanate EO phase shifters while EO-LN refers to monolithic Lithium Niobate on insulator waveguides which provide native EO switching capabilities.

longer sampling time. Consequently, the EO platforms require more cryogenic energy, making negligible the benefits of their high switching speed and low static power consumption. Detailed specifications of the different mesh platforms can be found in Table 12. Reducing coupling and component losses of EO platforms could reduce sampling time such that they operate in a regime where energy efficiency is differentiated by static power costs of the chips. Within this regime, where sampling time still dominates reconfiguration time, minimizing mesh static power strongly favors lumped element (LE) EO phase shifters over TO and traveling-wave (TW) EO. As photon sources, chip losses and detectors improve further and sampling time falls further, dynamic power and switching rate will matter more. If the sampling times fall to the point where reconfiguration rates on the scales of 10-100 GHz become essential, TW approaches may become practical and necessary as they can minimize dynamic energy and maximize throughput. The discussion in Appendix C, includes more technical details regarding the specifications and tradeoffs of these technologies.

We highlight that we took a conservative approach in the computation of the sampling time: we imposed that all of the input photons should be detected, yielding the exponential factor $\eta(2N_q)^{N_q}$ in (22), which implies a poor scaling of the energy efficiency with the system size. However, the energy efficiency could be significantly improved using loss-mitigation strategies [110], which are in fact routinely used in photonic quantum computing.

Moreover, although linear optical quantum computing constitutes one of the first schemes for universal quantum computing, the scalability issue has motivated the development of alternative architectures to realize fault-tolerant photonic quantum computing, such as fusion-based quantum computing [97] and the hybrid spin-optical architecture [111]. In addition, in the current era of NISQ photonic quantum

computing, photon native schemes [112] are being developed which are not necessarily gate based, in particular in quantum machine learning [113], photonic quantum reservoir computing [114, 115, 116], photonic quantum neural networks [117, 118]. An example of native task for NISQ photonic quantum computers is boson sampling [119], which has been extensively used as a benchmark of quantum advantage [120, 121, 122, 123, 124], and can be exploited to solve certain problems, such as graph problems [125], classification tasks [126, 116], or in hybrid classical-quantum algorithms [127]. The question of the energy efficiency of these alternative schemes is still poorly known, although it is generating an increasing interest, and has motivated recent work on an in depth investigation of the energy efficiency of boson sampling [109].

8 Future Directions

8.1 Error correction and fault tolerant quantum computing

Noise in quantum computing systems prevents the successful implementation of large-scale algorithms with application to real-world problems. Fully addressing this limitation requires the use of quantum error correcting codes (QEC) and, more generally, fault-tolerant quantum computing. However, moving to a fault-tolerant regime leads to a substantial increase in resource requirements, both in terms of hardware and energy consumption.

Replacing physical qubits by logical qubits implies a huge increase in the amount of resources, inversely proportional to the code encoding rate *i.e.* the ratio between the number of logical qubits per physical qubits. As an example, for a distance $d = 30$ surface code, the amount of physical qubits increases by a factor of roughly 1000. This overhead arises not only from the large number of physical qubits needed to encode logical qubits, but also from the implementation of fault-tolerant logical gates.

Moreover, the set of transversal gates is code dependent and cannot be universal [128]. Other techniques such as lattice surgery [129] or magic state distillation [130], which achieve a universal set of gates, require significant additional resources. Some logical operations such as lattice surgery also require a time which scales with the code distance d , often involving $O(d)$ physical operations for a single logical gate.

In addition, decoding, the process of identifying and correcting errors from measured syndromes, represents another major source of computational and energetic cost. Optimal decoding is usually an NP-hard problem, and could thus only be approximated. This entails a trade-off between the accuracy and the speed (or algorithmic complexity) of practical decoders. Therefore, efficient decoding demands advanced classical algorithms, which may themselves consume considerable computational power and prolong computation times. A realistic assessment of the total energy cost of fault-tolerant quantum computing must therefore include this classical contribution, which will certainly prove to be non-negligible.

Recent studies highlight the scale of these challenges. For instance, factoring a 2048-bit RSA integer using a surface-code-based fault-tolerant quantum computer has been estimated to require about one million physical qubits and five days of computation, assuming standard circuit-level noise and two-dimensional qubit connectivity [131]. This setup is compatible with superconducting qubit architectures, but the cooling power hosting this number of physical qubits is significant and there are existing trade-offs between noise and cooling temperature that were already investigated in this setting [3]. Other hardware platforms with different connectivity patterns may lead to different resource trade-offs [132]. Systems with higher qubit connectivity could, in principle, support quantum LDPC codes with higher encoding rates than surface codes, reducing the number of required qubits [133]. However, in these systems, implementing fault-tolerant logical gates remains difficult.

As of today, there is a challenge in listing the hardware components involved in fault-tolerant quantum computers, as only small scale examples exist. Simply scaling up the number of components of current computing architectures to millions of qubits leads to huge energy expenditure, which makes clear the necessary engineering and fundamental work that still needs to be done. Moreover, as explained above, syndrome decoding in QEC implies additional classical servers whose power consumption is also evolving.

We believe nonetheless that our work can be adapted to the case of fault-tolerant quantum computing. Once a hardware list of a fault tolerant quantum computer is established, QEC can be treated as any other

circuit applied to physical qubits, increasing the time where the hardware is consuming energy. QEC increases the number of gates that are applied to the qubits and thus increases the depth of quantum circuits implementing algorithms. As an example on a superconducting quantum computer, Equation 12 can be adapted to the case of circuits running on error corrected qubits. Defining a clock time is even more coherent for QEC codes, since each gate requires a set of operations that is consistent, even for different logical gates. On the other hand, QEC codes can also reduce the number of shots necessary to remove uncertainty due to noise on the qubits, which might reduce the total execution time of algorithms. They then introduce a trade-off on the execution time between depth increase and sample reduction. This does not apply, however, to the number of shots that are necessary because of the algorithm itself.

Taking all these aspects into account is essential for understanding the overall energy requirements of large-scale quantum computing. A complete analysis should thus consider the energy consumption associated with both quantum hardware operations and classical decoding processes. Since the field of QEC is going through very rich developments, both theoretical and experimental, we leave for future investigation the computation of the depth of those circuits.

8.2 Other computing paradigms and high performance computing

Our study is primarily focused on circuit-based computing, where algorithms are compiled into quantum circuits that are encoded in quantum hardware. We believe it can be adapted to new hardware by studying a baseline hardware architecture and a compilation model taking into account the specifics of the new platform. There also exist other computing paradigms that were not covered in this study, which could be adapted fit in our model. These are left as future research directions, with some initial thoughts on how to benchmark their energy consumption.

As an example, our methodology could be extended to measurement-based quantum computing (MBQC) [93], where algorithms are encoded in a series of adaptive measurements applied on universal graph states (also called cluster states or resource states). This can be especially adapted to the case of blind and private delegated computing [134, 135, 136, 137]. Benchmarking the energy consumption of MBQC in the context of delegated computation over a quantum communication network could give great insights on the cost of quantum cloud computing. In order to do so, one would need to analyze the hardware involved in maintaining highly entangled clusters of qubits, the compilation procedure behind MBQC, as well as the cost involved in back-and-forth communication over a classical or quantum channel.

An interesting photonic-based approach towards fault-tolerant computation is fusion-based quantum computing [97], where algorithms are also encoded in measurements of photonic cluster states. This approach being primarily pursued in the private sector, we were not able to get enough insights on the current devices to properly benchmark their energy consumption. The inherently probabilistic nature of quantum optics using state-of-the-art components suggests however a very consequent energy consumption devoted to cooling down the single-photon sources and detectors.

Another paradigm of interest is adiabatic quantum computing [138], based on quantum annealing [139, 140], where a computation is decomposed into a slow continuous transformation of an initial Hamiltonian into a final Hamiltonian, whose ground states contain the solution. It is of particular interest in the context of solving complex combinatorial optimization problems by finding the ground state of a system. This approach requires designing a Hamiltonian system and letting it evolve in a controlled environment. While it may be highly specialized in certain tasks, adiabatic computing could turn out to be more energy efficient than universal quantum computers in some cases, as less qubit control components are required. A comparison of the energy efficiency of adiabatic and circuit-based approaches solving the same task would be of great interest to understand which is the optimal design towards useful and sustainable quantum devices.

Another research direction is to benchmark the energy consumption of quantum machine learning techniques. Besides quantum neural networks [141, 142], this field also includes other approaches in development like quantum reservoir computing [143, 114, 144] or neuromorphic quantum computing [145]. To fit in the spirit of our model, benchmarking the energy efficiency of quantum machine learning requires defining a performance metric for a learning procedure (*e.g.* accuracy of the output) and understanding the hardware involved in the training and use of a quantum neural network.

Finally, a relevant research direction is to study how quantum computing would combine with high-performance computing (HPC). It is believed that quantum processors can be embedded into classical servers to enhance their ability to solve certain problems. A strong synergy between highly-advanced classical devices and emerging quantum platforms could prove to be the fastest way towards quantum-enhanced computing capabilities for research and society. A dedicated study of the energy efficiency of hybrid quantum-HPC machines would thus be of great interest.

9 Concluding remarks

In this work, we have developed a method for benchmarking the energy efficiency of quantum computers and showed how it applies to five of the leading hardware platforms currently envisioned. It quickly appears that one precise energy model adapted to all platform is not obvious and that it is safer to adapt a general metric, such as the energy efficiency presented in Eq. 1, to the specifics of each quantum computer architecture. We presented nonetheless different methods to compute a value of the energy efficiency for each platform, taking into account platform specification such as qubit connectivity, native gate, number of interaction zone or probabilistic operations.

With the insights of a diverse author list, ranging from academia to industry, we were able to investigate several state-of-the-art hardware architectures and to derive the power consumption of their different components (see Appendix A). This allowed estimations of the order of magnitude of the current energy consumption of quantum computers. While these numbers are related to specific technologies, it appears that most laboratories and companies actually use very similar state-of-the-art hardware from a few different vendors that have similar power consumption. As expected in the current state of quantum devices, maintaining the qubits in a controlled environment, either through cooling or maintaining the temperature of the room, is the most energy costly operation in most quantum computer architectures.

The core of this work resides in the compilation model (*i.e.* the translation of an algorithm in a hardware-aware circuit) that was established for each platform in order to estimate the running time of quantum algorithms. Analyzing the functioning of devices is essential to properly benchmark their energy consumption taking into account their specificity. We compiled key time-related parameters on each platform, such as the initialization time, the measurement time and the clock time of current state-of-the-art hardware. These parameters are crucial to understand how fast quantum computers can complete a given task and thus compute their energy efficiency. We believe that, while these compilation models can be further detailed and improved, they allow to already provide a good estimate of the computing speed of NISQ quantum computers.

Scaling up the results to quantum computers with thousands or millions of qubits is a tricky endeavor, as there is a consensus that radical changes and innovation in the engineering of the hardware are required for a sustainable scale-up. It is challenging to draw a strict line between the number of qubits and the number of hardware elements. Clever ways of sustaining and manipulating more qubits with less hardware are yet to be discovered through fundamental research. It can be however expected that the cost of the qubit control electronics will quickly outgrow the cooling cost as the number of qubit grows. Many more qubits could theoretically fit in current environment control systems without significantly altering their energy cost. The current road taken by the market of quantum technologies, where technologies are evolving faster than fundamental research, is to fit as many qubits as possible repeating the same technology, no matter the cost. Moreover, the implementation of QEC codes will imply a significant power dedicated to classical decoding algorithms. The cost of control electronics and classical computing might thus rapidly grow in a significant way.

We would like to emphasize that the results of this study alone cannot be used directly to make a fair comparison between platforms, since they are at different stages of development. While it seems that superconducting-qubit computers have a higher energy efficiency, other platforms might be more efficient at resolving specific tasks. As we showed, the native gate set of a platform can impose very large overhead during the compilation of an algorithm into a quantum circuit. Moreover, the specifics of a device constrain the execution time of certain operations and favor some families of algorithms. Consequently, the time required for a specific task on different platforms varies greatly. On top of that,

as mentioned in the introduction, other aspects, such as the manufacturing cost of the hardware elements present in the quantum computer, might outgrow by several orders of magnitude the running cost of the device.

Our work can be used to guide engineering and policy-making decisions in the near future of quantum technologies. Our framework can readily be used to benchmark different materials, *e.g.* different chip materials for photonic computing, architectures, *e.g.* different qubit connectivity in solid-state computers, or techniques, *e.g.* continuous reloading of atoms in neutral-atom computers. In general, it appears that significant improvement in energy efficiency can originate from increasing computing speed without reducing the power demand of particular elements in the hardware. With clever experimental advances, the computing time can already be reduced by huge factors, while reducing the power demand of the majority of the hardware requires skills that are often out of reach for researchers in quantum technologies. This also emphasizes the necessity of building strong synergies between hardware constructors and researchers. We hope that our results will inspire more researchers to take on the task of benchmarking the energy of their device and guide our field towards a more sustainable future.

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Appendix

A Hardware parameters

In this appendix, we give the detailed list of hardware components, and operation times, used to estimate the energy efficiencies of all the quantum computing platforms and their architectural variations present in this work.

We report the number of components in the quantum computer architecture, their power consumption, and the references used to make our estimation. In general, hardware components can scale differently with the number of qubits N_q . Some components, such as the amplifiers for readout lines in a solid-state computer, scale linearly with the number of qubits. If possible, we explicit the scaling with N_q , should the overall design of the architecture remain the same as the number of qubit grow. However, there is no fundamental scaling law for the hardware components in quantum computers. A combination of fundamental and engineering work should improve these scaling laws as technology progresses.

On the other hand, there are hardware components that do not directly scale with the number of qubits. This is the case for example of cryostats or lasers that produce optical traps. Current cooling systems can host thousands of qubits and, should the system size grow beyond that point, one would

have to replace these components with bigger, more powerful ones. The scaling of the power consumption of these device is not a direct function of the number of qubits, as many companies are working on more efficient cooling systems. In this case, we specify the number of components reported in our observations on current experimental devices. When possible, the power consumption of each element was measured in real life implementations.

To account for scheduling and classical processing that may happen during the execution of quantum algorithms, classical servers are included in the component lists. In some of the current implementations, one or two computers have enough computational capabilities to handle the required operations. When this is the case, classical computers with a power consumption of 150W (which includes a screen) are added to the component list. In the future however, classical processing happening in parallel of quantum computation needs to be carefully studied, especially when considering syndrome decoding in quantum error correction codes.

In addition, we report notes on the energy consumption of cryogenic systems, since they are the most energy consuming hardware elements and they can be present in many quantum computer architectures.

A.1 Superconducting qubits

A.1.1 Baseline architecture

Table 1 describes a simplified reference architecture depending on the number of readout lines N_R , which can host approximately five qubits each, and considering a cryogenic setup and a qubit control cluster capable to host 50 qubits. Based on Qilimanjaro’s superconducting architecture, we survey a Qblox Cluster with 20 module slots [146]; how these slots are populated determines the available control and readout capacity. For example, up to 80 baseband control outputs with QCMs, 40 RF control outputs with qubit control RF modules (QCM-RF) IIs, or 20 RF readout lines with QRM-RFs. RF signal generation is integrated within the cluster rather than supplied by a separate RF source. The rated maximum power of the cluster is 1.1 kW, though experimental measurements have shown a steady-state consumption of approximately 600 W, with spikes of around 10 W during circuit execution; nevertheless, we use the stated max power consumption. Each readout line requires two room-temperature low-noise amplifiers (LNAs) for warm-side signal amplification, and one cryogenic HEMT amplifier at the 4 K stage for the first stage of qubit signal amplification [147]. The qubit control and readout electronics are integrated into a modular cluster. In our architecture, a single readout line serves approximately five qubits. The cryogenic system has been chosen according to CESGA’s QMIO computer, discussed in the following section. The compressor drives the pulse tube pre-cooler of the dilution refrigerator, while the Gas Handling System (GHS) manages helium circulation and vacuum pumping [148]. Some installations deploy a second GHS for redundancy and improved operational stability.

Component	Number	Power consumption (W)
Compressor	1	9500
Gas Handling System	1	2000
Chiller	1	4600
Qubit control and readout cluster	1	1100
RF Source	N_R	70
Room Temperature LNA	$2 \cdot N_R$	1.4
HEMT PBA	N_R	100
Server	1	800

Table 1: Main components found in a superconducting qubit based quantum computer with a single feed line and N_q qubits. As a reference, the average feed line can host 5 superconducting qubits. These values are based on components from both CESGA’s and Qilimanjaro’s quantum computers.

Table 2 reports average operation times observed in current superconducting quantum computers. Based on Qilimanjaro’s superconducting devices, we define the clock time as the duration of a native

two-qubit gate operation. Depending on the architecture and native gate being implemented, two-qubit entangling gates such as the CZ or iSWAP typically take between 30 and 200 ns [149]. For concreteness, we adopt $t_{\text{clock}} \approx 50$ ns as a representative value. We observed that measurements take approximately $t_{\text{meas}} \approx 1600$ ns, consistent with the range of hundreds of nanoseconds to a few microseconds reported in the literature [150]. Between each circuit sample, qubits can be reinitialized either passively, by thermalization with the cold environment, or actively, using circuit elements to extract the excited-state population. Passive reset requires waiting at least five to seven times the qubit relaxation time T_1 , adding a minimum of $50 - 500$ μs per cycle for typical transmon devices [146]. We observed $t_{\text{reset}}^{\text{pass}} \approx 200$ μs , while active reset reduces this to $t_{\text{reset}}^{\text{act}} \approx 5$ μs , a roughly two-order-of-magnitude improvement.

Parameter	Symbol	Value
Clock time	t_{clock}	$50 \cdot 10^{-9}$ s
Measurement time	t_{meas}	$1.6 \cdot 10^{-6}$ s
Active reset time between shots	$t_{\text{reset}}^{\text{act}}$	$200 \cdot 10^{-3}$ s
Passive reset time between shots	$t_{\text{reset}}^{\text{pass}}$	$5 \cdot 10^{-6}$ s

Table 2: Operation times for a superconducting quantum computer.

A.1.2 Example of CESGA’s QMIO computer

QMIO is a 32-qubit superconducting quantum processor based on coaxmon technology developed by Oxford Quantum Circuits and operated at the Galicia Supercomputing Center (CESGA). Unlike the reference model described above, QMIO represents a fully deployed system with measured steady-state power consumption at the facility level, including cryogenic infrastructure, control electronics and classical processing.

The comparison between the literature based superconducting model described above and the experimentally characterized QMIO 32-qubit processor reveals that, within the 50 qubits system size, the steady-state power consumption of these superconducting platforms remains confined to a relatively narrow interval dominated by cryogenic overhead. Specifically, the architecture discussed above, reaches 20 kW of power consumption, and the QMIO processor operates at 21.4 ± 2.4 kW steady-state (Table 3). The component-level power breakdown of QMIO is summarized in Table 3 and the measured power breakdown for QMIO is shown in Figure 18. It confirms that cooling alone accounts for roughly 76% of the total system power.

Component	Number	Power (W)	Notes
Compressor (pulse tube)	1	9000 ± 500	Derived from cryogenic increment (baseline subtraction method).
Chiller (water cooling)	1	4200 ± 400	Derived from cryogenic increment.
Gas Handling System (GHS)	1	1800 ± 200	Derived from cryogenic increment.
Cryostat auxiliaries (pumps, controllers)	1	1320 ± 200	Residual to close cryogenic increment budget.
Per line FPGA control/readout modules	32	125 ± 19	Allocated from measured control electronics (FPGA-per-qubit architecture).
RF/IF up- and down-conversion electronics	1	600 ± 100	Included in measured control electronics.
Clock distribution system (2 GHz differential)	1	Included	Described in QMIO architecture; included in control electronics power.
Control server and orchestration	1	440 ± 80	Included in measured control electronics.

Table 3: Main components found in CESGA’s QMIO superconducting quantum computer ($N_q = 32$) and their operational power consumption (steady-state). The power consumption of each element was directly measured on the existing setup. The error range corresponds to variations in the energy consumption over time.

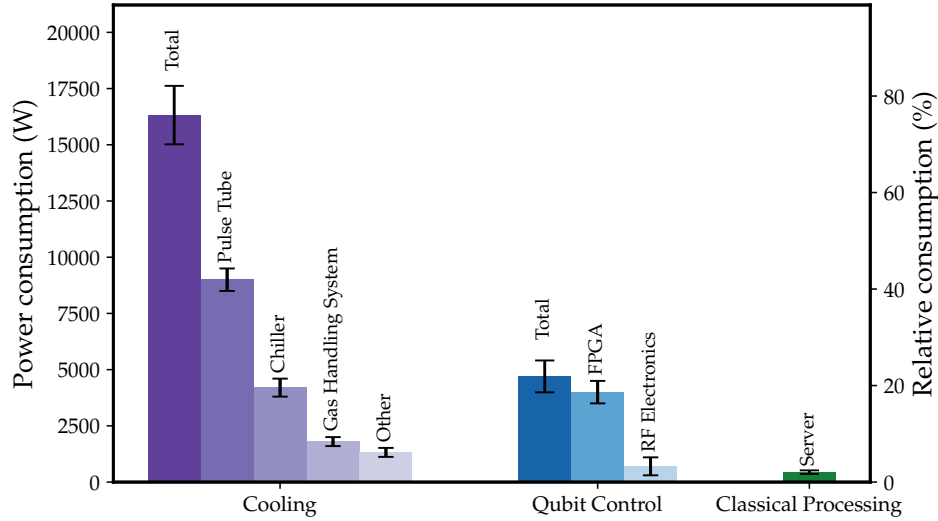


Figure 18: Energy breakdown of a QMIO superconducting-qubits computer (32 qubits), according to the components shown in Table 3. In the x axis, the components are grouped according to their purpose in the computer. Each group has a bar that corresponds to the total accumulated power of its components. The left y axis shows the absolute value of power consumption of the components, and the right y axis the relative value with respect to the overall consumption of the entire computer. Error bars correspond to variations in the power consumption over time.

A.2 Spin qubits

This appendix presents the architectures used in the main text to compute estimations of the energy efficiency of spin-qubit quantum computers. Table 4 describes a simplified reference component list, constructed having in mind a certain architecture requiring N_{DC} number of DC connections, N_{BB} number of baseband control for fast charge manipulation, N_{MW} number of microwave control line for qubit operations and necessitates N_R number of readout lines. Table 5 presents how these numbers relate to the number of qubits in the computer for three different architectures as well as their reset time between shots, clock time, and measurement time.

Component	Number	Power consumption (W)	Ref.
Pulse tube	1	9500	
Chiller	1	4600	
GHS	1	2000	
Classical computer	2	150	
DC control	N_{DC}	~ 0.5 per unit	[151]
Baseband control	N_{BB}	~ 7.5 per unit	[152]
MW control	N_{MW}	~ 24 per unit	[153]
Readout lockin	N_R	~ 48 per unit	[154]
Low-noise amplifiers	N_R	~ 0.1 per unit	[155]
Room-temperature amplifiers	N_R	~ 0.5 per unit	[156]
Magnet power supply	1	neg. in steady state	

Table 4: Main components found in a spin-based quantum computer

	2D ($N_q = n \times n$)	Linear ($N_q = n$)	2D crossbar ($2N_q = n \times n$)
N_{DC}	$3n^2 - 2n = 3N_q - 2\sqrt{N_q}$	$2n - 1 = 2N_q - 1$	$4n + 1 = 4\sqrt{2N_q} + 1$
N_{BB}	$3n^2 - 2n = 3N_q - 2\sqrt{N_q}$	$2n - 1 = 2N_q - 1$	$4n + 1 = 4\sqrt{2N_q} + 1$
N_{MW}	$n^2 = N_q$	$n^2 = N_q$	1
N_R	$n^2/2 = N_q/2$	$n/2 = N_q/2$	$n = \sqrt{2N_q}$
t_{reset} (ms)	0.1	0.1	0.1
t_{clock} (ns)	10	10	250
t_{meas} (ms)	0.01	0.01	$0.01 \times \sqrt{N_q/2}$

Table 5: Number of hardware components and operation times in a spin-qubit based quantum computer for the 3 different connectivity considered.

The prototypical example of a quantum register can be envisioned as a two-dimensional array of quantum dots arranged in a square lattice of size $n \times n$. The power consumption strongly depends on how this quantum register is controlled. In the worst-case scenario, each qubit and coupling gate is controlled individually (a full one-to-one control scheme). This would require one DC line per quantum dot and one per coupling gate, yielding approximately $N_{DC} \sim 2n^2$. The same reasoning applies to baseband control lines, giving $N_{BB} \sim 2n^2$. Microwave control is only required for qubit manipulation, leading to $N_{MW} \sim n^2$.

Regarding the readout, several options exist. Charge sensors typically complicate the geometry of the quantum-dot array and still need to be coupled to external resonators to achieve high bandwidth. We therefore restrict ourselves to gate-based dispersive readout, as this choice does not substantially affect the total power consumption. In the worst-case scenario, one would require a resonator per pair of qubits for parity-based readout, and one readout line per resonator, resulting in $N_R \sim n^2/2$.

A second strategy consists of reducing the number of control lines. For example, Ref. [54] proposed a crossbar approach, in which control lines are shared among rows, columns, and diagonals of the array. In

this case, instead of having a number of control lines that scales quadratically with the number of qubits, the scaling becomes linear in n , i.e. proportional to the square root of the total qubit number

A.3 Trapped ions

This appendix presents the architectures used in the main text to compute estimations on the energy efficiency of trapped ions quantum computers. Table 6 describes a simplified reference component list, based on the trapped ions device found in the NQCC laboratory. Here we consider a room temperature device, albeit the energy cost of operating at cryogenic temperature is also shown. This considers a processor running with a single ion species, and we split the control system into active and passive, where active means it's expected to respond in real time to perform gates. In this case, the active control system would be the only component expected to vary for the 5 and 10 traps computer. The considered computers can perform single- and two-qubit gates in parallel (in order to take full advantage of their multiple gate zones), thus we need these two active control systems that can target different sets of atoms and perform independent rotations on their state at the same time. However, there are two instances for which we expect the number of active control system to scale further. First, if one wants to perform more species of gates at the same time (such as different kinds of single-qubit gates), N_{control} would be equal to the number of independent gates to be executed in a single time step (with $N_{\text{zones}}^{\text{gate}}$ being the maximum). Then, if the number of gate zones is large enough, we expect that a single active control system would struggle to execute the same gate on all of them, as an electronic signal can only be split a given number of times before it becomes too small to be detected by the hardware. In that case, we expect N_{control} to scale with $N_{\text{zones}}^{\text{gate}}$.

Component	Number	Power consumption (W)
HVAC	1	7500
Gate drive	1	700
Laser system	1	825
Classical computer	1	150
Image system	1	100
Magnetic field generation	1	1
Active control system	N_{control}	100
Passive control system	1	200
Compressor*	1	7500
Chiller*	1	2000

Table 6: Main components found in a trapped ions based quantum computer. The power consumption values were directly observed on the trapped ion device in the NQCC in the UK. The components marked with a star (*) are not mandatory in QCCD trapped ions devices, and are commonly omitted. Although present in the NQCC laboratory, we removed them from the main analysis.

Table 7 shows the reset time between shots, clock time, measurement time and shuttling time of ions between storage and gate zones, for devices with 1, 5 and 10 gate zones. The reset time accounts for transport back to memory (or storage) zone, Doppler and sideband cooling. The clock time accounts for the average time to perform the two-qubit gates, plus any transport that might happen between gate zones. This consequently leads to a longer clock time for the multi-zone computers, due to extra transport needed to arrange the qubits. To estimate the transport time of the multi-zone devices, we assumed that gate zones are connected in a linear geometry, and set the upper bound to the time required to populate the furthest gate zone. Assuming that these devices can move ions safely at 3 m/s, and the distance between gate zones is 750 μm [72], we expect the clock time of multi-zone devices to be $t_{\text{clock}}^{\text{multi-zone}} \approx 250 \mu\text{s} \times (N_{\text{zones}}^{\text{gate}} - 1)$, to move the ions between the farthest apart gate zones. For the 5- and 10-gate-zones, this movement is the most time-consuming operation in a single circuit layer, and is thus considered to be the clock time. In contrast, the clock time of the single-gate-zone device corresponds to the additional transport to bring a pair of qubits to the same potential well (100 μs) and the time to

perform a gate itself ($70 \mu\text{s}$). This time is based on laser gates, but the fastest microwave gates would be performed in 300us . Since two qubit gates require more transport, and transport induces heating in the ions, cooling is normally performed in between gates [157]. For the sake of simplicity, we exclude this cooling time out of our model, as it would only represent a quantitative difference and not the overall conclusions of the model.

	$N_{\text{gate zones}} = 1$	$N_{\text{gate zones}} = 5$	$N_{\text{gate zones}} = 10$
N_{control}	1	2	2
t_{reset} (ms)	150	150	150
t_{clock} (μs)	170	1000	2250
t_{trans} (ms)	50	50	50
t_{meas} (ms)	0.5	0.5	0.5

Table 7: Number of active control systems and operational times in trapped ion quantum computers with 1, 5 and 10 gate zones.

The components detailed in this section correspond to a basic QCCD trapped-ions device build with a storage zone and several gate zones connected linearly. There are multiple modifications that can be made to further improve the performance and efficiency of such device. For instance, one could modify the distribution of the traps to define cache zones, or improve the connectivity between gate zones, which would optimize the transports needed during the execution of a circuit. These modifications, however would need to be combined with a powerful compiler to take advantage of the device’s geometry (this is the case of Quantinuum’s Helios computer [72]). In that sense, we expect state-of-the-art and near-future devices to display higher power demands, but within the same order of magnitude as our estimations. However, if this resources are used properly, the computing time of such computers would be significantly optimized and could overcompensate the extra power consumption, delivering higher energy efficiency.

A.4 Neutral Atoms

This appendix presents the architectures used in the main text to compute estimations on the energy efficiency of neutral atoms quantum computers. More specifically, we base our estimation on the rubidium-87 neutral atom processor hosted in Harvard, where digital entangling operations are performed via the $n = 53$ Rydberg state. The system itself supports up to a thousand individually trapped atoms with coherent transport performed in parallel on over 100 atoms at a time. In this system, atoms can be re-used and lost atoms can be refilled from a reservoir to decrease cycle time through state-selective loss-detecting lattice readout. The continuous array system that we include in our analysis is based on Ref. [87] where atoms are reloaded with states up to 30,000 qubits per second. Table 8 describes a simplified reference component list, implementations that can host up to a thousand traps and $N_q = 400$ qubits. The table include hardware components for the baseline neutral-atoms architecture based on the Harvard computer, and the extra components required to implement continuous reloading.

Table 9 shows the relevant operational times of the studied neutral-atoms computer. These estimations are based on the processor operating with qubit reuse as presented in [87]. Here, atoms are loaded and cooled in a magneto-optical trap and then filled into tweezers, consisting of the computational array as well as a atom reservoir. This process is the most time intensive and required approximately 500ms. In order to increase cycle rates, the atoms are not discarded after an experimental shot, but instead, lost atoms are refilled from a reservoir and qubits are reinitialized. In contrast to MOT reloading, this process takes on the order of 10 ms for each shot, which corresponds to the t_{reset} considered in this work. Atoms can be refilled from a reservoir until it is depleted, therefore, it is highly dependent on the array and reservoir size until atoms must be reloaded from the MOT. For the purposes of this calculation, we estimate reloading every 2400 gate layers. For the sake of simplicity, we assume that reloading can be performed *during* a circuit execution, not only between samples. In the case of a unitary circuit that would require to copy the state of the qubits before reloading, which we do not account for. Transport and reconfiguration times are also given as broad estimates that are highly dependent on the circuit structure.

Component	Number	Power consumption per unit (W)	Ref.
Trap lasers	1	1500	[158]
Lasers for Rydberg gates	1	1900	[159, 160]
Other lasers	2	900	
	7	150	
Magnetic field generation	1	1050	
Camera	1	155	[161]
Vacuum pump	1	100	
Classical computers	1	150	
HVAC	1	7500	
Other electronics	1	2000	
Laser Chiller (small)	5	600	[162]
Laser Chiller (large)	2	1000	[163]
Extra components for continuous reloading			
Transport laser	1	350	
Laser Chiller	1	1500	
Additional Vacuum pumps	1	700	

Table 8: Main components found in a neutral-atoms quantum computer. The values without reference were directly observed on the experimental setup found at the Harvard laboratory [78].

Moving atoms in optical tweezers is generally limited by atom loss and heating to approximately 1 ms [85]. Generally, the arrays are around 200 μm in extent and therefore, considering trap transfer times as well, we estimate that both reconfiguration of the atoms during successive gates as well as moves from the storage region take on the order of 500 μs . Considering that the reconfiguration of the atoms in the interaction is a process that takes place before most of the layers of the circuit, and that single- and two-qubit gates take the scale of 10 μs and 150 μs , respectively, we assume $t_{\text{clock}} = t_{\text{reconfig}}$. On the contrary, transport from the storage to the interaction-zone is a less common operation, characterized by the β_{trans} parameter. Finally, we have chosen the measurement time to be ≈ 10 ms. Again this is highly dependent on readout method, and new schemes have been shown readout times on the microsecond scale [164].

	Periodic Reload	Continuous Reload
t_{reset} (ms)	10	-
t_{clock} (μs)	500	500
t_{trans} (μs)	500	500
t_{meas} (ms)	10	10
t_{reload} (ms)	500	-
Reload freq. (# of layers)	2400	

Table 9: Extra hardware components and operational times of two neutral-atoms devices, with and without the ability to perform continuous reload of the register during computations.

A.5 Photonic architectures

This appendix presents the architectures used in the main text to compute estimations on the energy efficiency of photonic quantum computers. Table 10 describes a simplified reference component list, based on real life implementations of chip-based photonic computation, in particular the one from Quandela [98]. It assumes cryostats that can host up to one quantum dot source and 25 superconducting nanowire single photon detectors (SNSPDs) as well as demultiplexer able to send at most 12 photons on the chip per source. Table 11 and 12 show the efficiencies of components that are used to compute the end-to-end

transmission of devices using different chip technologies. More information about the chip technologies can be found in Appendix C.

Component	Number	Power consumption (W)	Ref.
Laser	1	4	
Cryogenic system for the Quantum dot and the SNSPDs	$\left\lceil \frac{2N_q}{25} \right\rceil$	1500	
Peltier cooling of the chip	1	200-300	
Demultiplexer system (DMX-12)	$\left\lceil \frac{N_q}{12} \right\rceil$	~ 50 per DMX	
Classical control electronics (FPGA, DAC)	$\left\lceil \frac{4N_q^2}{16} \right\rceil$	~ 15 W	[165]
Classical computers		150	

Table 10: Main components found in a single photon source photonic quantum computer with N_q qubits.

Name	Symbol	Value
Source efficiency	η_{of}	71.2%
Detector efficiency	η_d	98%
Demultiplexer efficiency	η_{dmx}	83%

Table 11: Component efficiencies in photonic computers.

Chip technology	Glass mesh	EO-LN	EO-BTO
Power consumption of the chip elements	50-100 mW	60 μ W	121 nW
Time to reconfigure the phase shifters	1 μ s	44 ps	145 ps
Coupling loss at in/output of the chip	0.5 dB/Facet	3.4 dB/Facet	.127 dB/Facet
Mach-Zehnder interferometer loss	0.1 dB/MZI	0.15 dB/MZI	.21 dB/MZI
Propagation loss	0.3 dB/cm	0.025 dB/cm	0.016 dB/cm
Ref.	[166]	[167, 168]	[169, 170]

Table 12: Differences between the chip technologies considered in photonic computation.

A.6 Cooling systems

Cryogenics are a very important part of many quantum computing architectures. Since the quantum systems in which information is encoded need to be protected from decoherence and outside influences, it is sometimes necessary to cool them down to temperatures below 20 mK. Moreover, some technologies such as single photon detectors have a higher efficiency at low temperature (around 4K). Pulse tube, water cooler and gas handling systems are a crucial parts of superconducting, spin and photonic quantum computers. Unsurprisingly, they represent one of the main source of energy consumption in the total energy balance of these platforms.

In a typical dilution fridge used for quantum computing uses different Helium isotopes and cooling happens in two stages. A first phase of pre-cooling down uses a so-called pulse tube to cool down a volume up to ≈ 4.2 K. This is the maximum achievable due to physical constraints: below this temperature gas density drops. To reach a lower temperature, a dilution cycle of $^3\text{He}/^4\text{He}$ starts, using an secondary pulse tube. This allows systems to reach temperatures < 10 mK. To understand the energy consumption of this technology, we use, as a reference, a cryogenic refrigerator used in our lab in ICFO, Spain [171]. We

note, however, that there are few companies that produce this technology and the values that we show in the different tables of this appendix are compatible with all current state-of-the-art dilution fridges.

We observed that most of the energy consumption stems from the primary pulse tube. For example, measurements performed with a voltmeter on the cryogenic refrigerator [171] revealed that the consumption of the secondary mechanism, used to perform the $^3\text{He}/^4\text{He}$ cycle, has a power consumption of 200 W while the primary pulse tube has a power consumption around 8000 W. Moreover, the energy spent by the fridge depends on two main factors: the heat production of the system inside the fridge and the volume to cool down. It is challenging, and out of the scope of this study, to derive a general relation between dissipated heat, volume and power consumption, as done for example in [3]. In this work, we use the power consumption observed in real-life implementations of solid-state quantum computers using dilution fridges.

Since the volume and amount of control electronics should increase significantly as the number of qubits grows, the energy spent in cooling can only increase. Our observations show however that the energy spent in the electronic components required to address the qubits might outgrow the cooling energy in the future.

B Compilation of algorithms

To run on a quantum computer, a quantum algorithm needs to be compiled into a sequence of gates that can be run efficiently on the specific hardware used. The gates are scheduled in time steps, where the gates that do not act on the same qubits can be performed in parallel. The number of time steps defines the depth of the circuit. The circuit depth of an algorithm is central to our study, since it defines how long an algorithm will take to be ran on a quantum computer.

B.1 Solid-state qubits

Given the qubit connectivity in a solid-state platform, this compilation process can result in very different depths for the same algorithm. In this section, we discuss the influence of the compilation method and the qubit connectivity of a platform on the circuit depth of an algorithm.

In order to characterize the depth overhead of an algorithm executed on a solid-state processor with given connectivity, we need to account for the distance between qubits that interact together. The connectivity between qubits is represented by a graph $G = (V, E)$ where each vertex $v \in V$ corresponds to a qubit and, if two vertices are connected by an edge $(i, j) \in E$, we can perform a physical two-qubit gate between them. In order to solve this issue, one has to include SWAP gates that place qubit states in adjacent vertices connected with an edge. This process is called routing. For instance in a linear graph, a CNOT gate between adjacent qubits does not result in any overhead. On the other hand, a CNOT gate between qubits that are 4 nodes apart, will add 2 extra layers of SWAP gates and relabeling (see Figure 19).

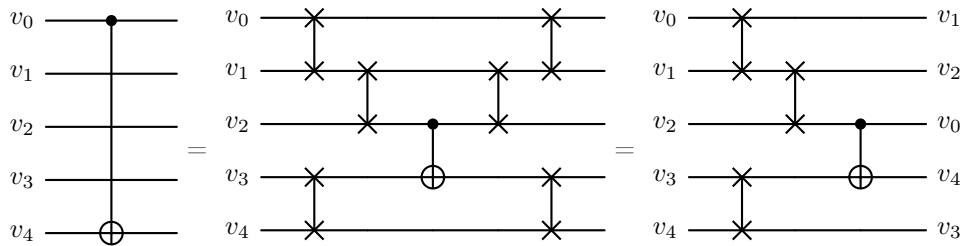


Figure 19: Implementation of a CNOT gate between v_0 and v_4 , on a quantum system with only line connectivity (with qubit connectivity between v_i and v_{i+1}). The last two layers of the middle circuit are optional and can be replaced by qubit relabeling.

In a general graph G , we can define the Manhattan distance $d(v, v')$ which corresponds to the length of the shortest path between two vertices in G . We can then replace a CNOT between two distant qubits v, v' by $d(v, v') - 1$ SWAP gates and 1 CNOT gate. Considering that each SWAP gate is done by adding 3 layers of CNOT gates between adjacent qubits, this implies a total overhead of $3 \times \left\lceil \frac{d(v, v') - 1}{2} \right\rceil$ layers. In order to generalize this expression for a given graph, we use the average shortest path between all pairs of nodes of a graph G :

$$\overline{d(G)} = \frac{2}{|V|(|V| - 1)} \sum_{v \neq v' \in V} d(v, v'). \quad (24)$$

Therefore, we estimate that the average depth overhead needed to connect any pair of qubits in a processor with connectivity graph G is $3 \cdot \left\lceil \frac{\overline{d(G)} - 1}{2} \right\rceil$.

Furthermore, we assume that the gate overhead that connect two different pairs of qubits can also be parallelized. Therefore, we assume that a layer with only one two-qubit gates will produce the same overhead as a layer with many two-qubit gates. With this in mind, the depth overhead of a circuit depends on the number of layers that have (any number of) two-qubit gates D^{2q} , as $D^{2q} \cdot 3 \cdot \left\lceil \frac{\overline{d(G)} - 1}{2} \right\rceil$.

Let $\alpha_{2q} = D^{2q}/D$ be the fraction of layers which include a two-qubit gate in the transpiled circuit. The circuit depth after routing is given by:

$$D \approx D_0 \left(1 + \alpha_{2q} \cdot 3 \cdot \left\lceil \frac{\overline{d(G)} - 1}{2} \right\rceil \right) \quad (25)$$

An upper bound of this value can be obtained by substituting the average shortest path length $\overline{d(G)}$ with the diameter of the circuit $d(G)$, which represents the [longest] shortest path length between nodes. While this model can be improved in future studies, we believe this allows us to provide a fair order of magnitude of the energy efficiency of solid-state platform. Our compilation model has the benefit of being simple and easily integrable into the larger model to benchmark the energetic cost of quantum computing.

In this work, we used the `rustworkx` python library to represent different connectivity graphs, and compute $\overline{d(G)}$ exactly for each of them. The graphs that we consider the most relevant for solid-state computing are:

- **Fully Connected Graphs:** In these graphs, all pairs of nodes are connected with an edge. Rather than an actual implementation for solid-state computers, fully connected graphs correspond to an ideal-case scenario in which there is no connectivity constraints at all. This serves us to compare the overhead produced by other constrained graphs with respect to the ideal case, where the circuit does not require any gates. For fully connected graphs, $\overline{d(G)} = d(G) = 1$.
- **Linear Graphs:** These graphs correspond to a one-dimensional string of qubits (nodes) connected to their nearest neighbor. This is the most basic form of connectivity that allows all qubits to be able to interact between them, since there is always a path that connects any pair. The shortest paths between nodes of a linear graph can be reduced by connecting the two ends of the graph. This would correspond to a Circular Graph, that we also include in Figure 2. For linear graphs, $\overline{d(G)}, d(G) \propto N_q$.
- **2D Lattice Graphs:** In general, 2D Lattice Graphs, are the current state-of-the-art approach for scaling solid-state quantum computers. There are several sub-groups of 2D graphs, based on the geometry of the lattice cells. The most common geometries, and the ones that we include in this work, are square lattices and heavy-hex lattices (graphs iii and iv displayed in Panel (b) of Figure 20). These graphs are often optimized for quantum error correction codes [172, 173].

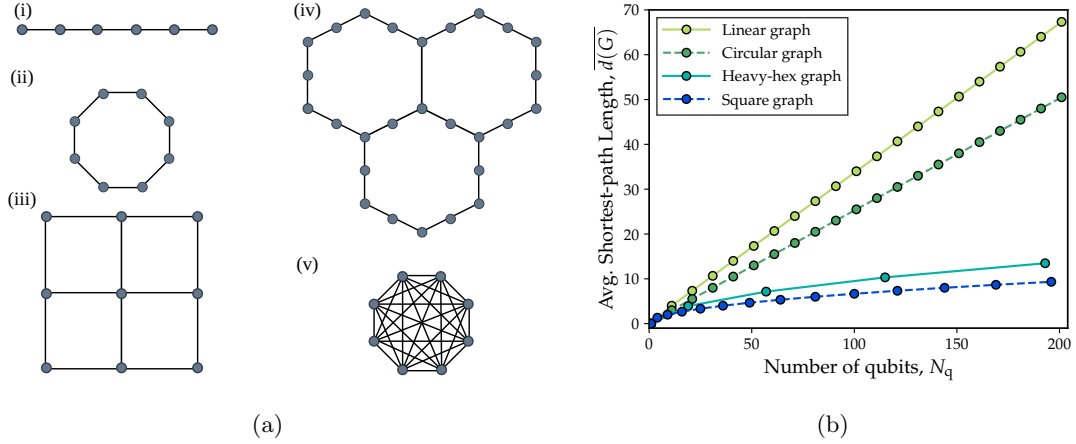


Figure 20: Different types of connectivity considered in this work for solid-state-qubits computers. Panel (a) displays the shape of the following connectivity graphs: i) Linear, ii) Circular, iii) Square-lattice, iv) Heavy-hex and v) Fully-connected. Panel (b) shows how the average shortest-path length $\overline{d(G)}$ of these graphs (excluding the fully-connected one, for which $\overline{d(G)} = 0$) scale with the number of qubits.

B.2 Atom-based qubits

After the transpilation step of an algorithm into a quantum circuit of depth D_0 , the routing step of the compilation process ensures that gates can be applied to the appropriate qubits. Although it is usually said that atom-based devices have all-to-all connectivity, they still have major parallelization constraints. In general, in an atom-based device the gates can only be performed in a limited number of interaction zones, N_{zones} , which limits the number of gates that can be executed in parallel. In this work, we propose a simple characterization of circuit compilation for atom-based computers, in order to estimate the extra time taken by devices that have different N_{zones} .

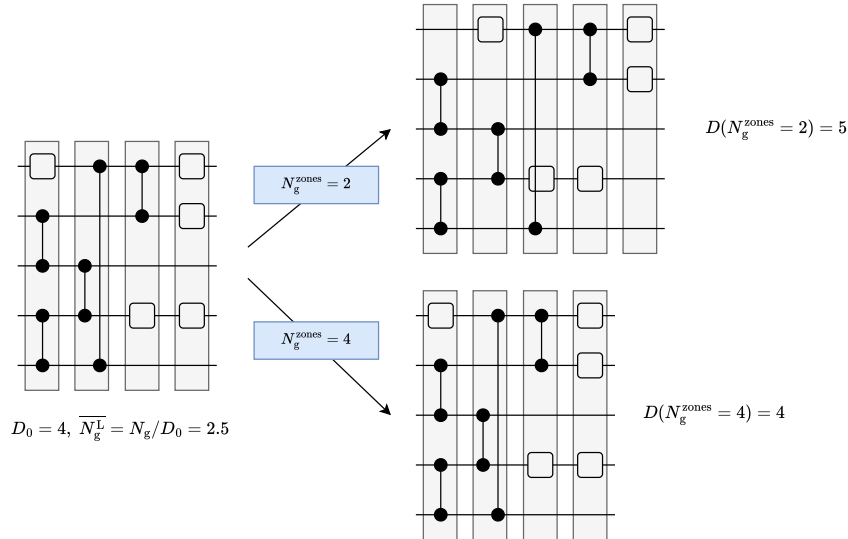


Figure 21: Diagram of a circuit being compiled into two atom-based devices with 2 and 4 gate zones. The horizontal lines represent different qubits, the small squares represent single-qubit gates, and the vertical lines represent two-qubit gates. The layers of the circuits are highlighted with gray vertical rectangles.

In order to execute a quantum circuit on a given device, the number of gates in each layers of the circuit is limited by the number of gate zones. Therefore, all the layers containing more than N_{zones} gates have to be split into more layers, which introduces a depth overhead. Assuming that large circuits tend to have an even distribution of gates per layer, we can approximate the number of times this happens by using the average number of gates per layer:

$$\overline{N_g^L} = \frac{N_g}{D_0}, \quad (26)$$

where N_g is the total number of gates in the circuit. If the number of gates in the layers is lower than the number of gate zones, the circuit can be executed in the device without adding any extra layers. On the contrary, if there are more gates per layer than the available gate zones, the layers of the circuit will be split into smaller layers with less gates. Figure 21 shows a diagram of a circuit being compiled into a 2- and a 4-gate-zones device. With this reasoning in mind, the final depth of the circuit reads:

$$D = \left\lceil \frac{\overline{N_g^L}}{N_{\text{zones}}} \right\rceil \cdot D_0. \quad (27)$$

It is worth mentioning that the final depth of a real circuit could be further optimized for a particular device. For instance, let's say one wants to execute a circuit that has 7 gates per layer on a device with 5 gate zones. According to our description, each layer will be split in two. However, on many cases, the 2 gates that exceed the number of gate zones in some of the layers could be put together to compress the excess overhead. In addition, some circuits might have a large variance in gates per layer, in which case the depth overhead would be uneven through the circuit. However, as the number of gates zones increase, $\overline{N_g^L}$ variations also become less relevant, *e.g.* a layer with 21 gates and 39 gates will be both split in two for a device with $N_{\text{zones}} = 20$. For the sake of generality, we assume that the final depth of the circuit will not change in orders of magnitude from our estimations, but we want to stress that architecture-aware compilation processes can provide a certain degree of optimization on the execution time of circuits.

The computing time of a trapped-ions based quantum computer is highly influenced by the shuttling time of ions from zones in the device. In particular, we assume that between all layers, the device has to reconfigure the location of the ions in the traps that belong to the gate zones, and after a certain number of layers, the ions in the traps have to be exchanged for other ions held in the storage zone. The frequency of storage-to-gate-zone transport depends on the structure of the circuit, the number of qubits being used and the number of gate zones in the device. For instance, a circuit that has several consecutive layers with gates on the same set of qubits, will need less transport than a sparse circuit where most qubits are used sequentially. To characterize the ratio of layers that require ion transport, we define the parameter $\beta_{\text{trans}} = N_{\text{trans}}/D$, where N_{trans} is the total number of storage-to-gate-zone transportation. Therefore, the time to execute a circuit in an atom-based computer can be written as follows:

$$t_{\text{circuit}} = t_{\text{reset}} + D \cdot t_{\text{clock}} + \beta_{\text{trans}} \cdot D \cdot t_{\text{transport}} + t_{\text{meas}}. \quad (28)$$

According to our previous reasoning, $\beta_{\text{transport}}$ depends on the structure of the circuit, the number of qubits, and the number of gate zones of the device. For a given circuit, we expect that increasing the number of gate zones by a certain proportion, reduces the number of transports with, at least, the same proportion. For instance, a circuit that has to perform 100 transports in a device with 1 gate zone, will need, at most, 10 transports in a 10-gate-zones device. In that case, the ratio $\beta_{\text{transport}}$ will be 10 times smaller for the 10-gate-zones device. However, the relation between N_{trans} , $\beta_{\text{transport}}$ and the number of gate zones is not always so clear. Let's take a circuit that executes a large number of gates on a set of 10 qubits, and then performs another large number of gates on a different set of 10 qubits. In a 10-gate-zones device, this circuit needs only one transport and, for a big enough depth, has $\beta_{\text{transport}} \sim 0$.

Finally, there is a clear lower bound of $\beta_{\text{transport}}$ for any circuit with average gates per layer $\overline{N_g^L}$, executed on a device with N_{zones} gate zones. In a single layer of the original circuit, all the gates are applied to different qubits. Therefore, if $N_{\text{zones}} < \overline{N_g^L}$, the original layers will be split into new layers

that use different qubits, with no other option than having to transport them from the storage to the gate zones and vice versa. The minimum ratio of transports per layer is then:

$$\beta_{\min}(\overline{N_g^L}, N_{\text{zones}}) = 1 - \frac{1}{\left\lceil \frac{\overline{N_g^L}}{N_{\text{zones}}} \right\rceil}. \quad (29)$$

C Chip technologies for photonic computing

Programmable photonic chips form the backbone of photonic quantum computing platforms by providing the medium through which gates can be applied to quantum states. Such chips require interconnected waveguides for confining and directing photons to provide a set of orthogonal spatial modes as well as methods of dynamically tuning the coupling strength between these modes. Generally this is accomplished by means of Mach-Zehnder interferometer meshes where mode coupling is tuned by means of electronically controlled phase shifting elements. Different combinations of waveguide and coupling technologies have been mixed and matched to combine the advantages and mitigate the shortcomings of each. In this review we compare examples of these platforms limiting our scope to the technologies which dominate the current literature and industrial efforts. For the purposes of our analysis in Figure 17 we compare only those examples which have been explicitly integrated into or designed for programmable interferometer meshes.

A fair comparison between platforms must include electrical power to operate the phase shifters as well as coupling/propagation losses that change the time required to obtain samples, and switching speed, which sets gate initialization overhead. The phase-shifter power is the sum of static and dynamic components where static power maintains a setting and dynamic power updates it. When sampling time is long compared to initialization, the dynamic contribution can be negligible; in large meshes and long sampling windows, static power can dominate, so we focus on it.

Glass-based meshes are compelling for quantum photonics: excess loss as low as 4.35 dB fiber-to-fiber in a 24-mode mesh, fiber coupling losses .5 dB/facet, propagation loss below 0.3 dB/cm, 0.1 dB per MZI unit, polarization invariance, and quasi-3D waveguide fabrication [166]. The key limitation is that Silicon lacks native optical activity, so phase control relies on either electrically-driven plasma dispersion effects or Joule microheaters using the thermo-optic effect. Plasma dispersion phase shifters induce large and phase-dependent propagation losses making them unsuitable for quantum computing [174]. This constrains switching bandwidth to <1 kHz and creates thermal cross-talk that can require active stabilization, leading to ~ 2 s algorithm update time. Typical heaters consume 10–100 mW static power [166, 175]. Multiplying by the quadratic scaling of tuners in a Clements mesh and integrating over long sampling times can yield significant static energy.

Electro optic (EO) platforms have been the subject of intense development largely driven by telecoms. They use the Pockels effect to induce phase shifts linearly proportional to an applied electric field. The leading EO materials are lithium niobate (LN) and barium titanate (BTO). Both offer high index and wide transparency $n \sim 2.2$, $0.35 \sim 5 \mu\text{m}$ for LN and $n \sim 2.4$, $0.35 \sim 5 \mu\text{m}$ for BTO [176]. Their EO coefficients $-r_{33} \approx 26.9$ pm/V for LN and 125 pm/V for BTO – enable compact, efficient phase shifters [177]. These phase shifters can be either implemented monolithically or by evanescent coupling with thin films of EO material deposited onto low-loss Silicon-based waveguides [178]. Such heterogeneous integration allows the advantages of multiple materials to be combined such as the ultra-low loss of Silicon nitride (SiN) waveguides with the lower energy switching of BTO [169]. Device loss, bandwidth, and voltage depend on intricately coupled waveguide/electrode design beyond the scope of this discussion but two dominant device types are traveling-wave (TW) and lumped-element (LE).

In TW devices, electrodes form a transmission line so the RF wave co-propagates with the optical carrier, maximizing overlap and achieving modulation bandwidths $> 100\text{GHz}$. [179, 180, 181]. However, holding a setting requires continuous RF delivery. Even efficient TW designs can incur tens of mW, making them unattractive when sampling times greatly exceed switching times. In LE designs, the device is effectively a capacitor (EO layer as dielectric). Energy is mainly expended dynamically during updates; static consumption is near-zero aside from leakage (often $\sim 100\text{s}$ of nW). State-of-the-art LE LN

reports 15 GHz switching [182, 183]. At 1 kHz sampling, typical per-device dynamic power can be 1 μ W.

EO phase-shifters are promising for scalable photonic QC, with the main barrier being industrially scalable heterogeneous integration processes; LN is currently most mature, while BTO is advancing rapidly [169, 184, 170]. EO platforms may pay higher insertion-loss penalties: LE meshes in monolithic LN report 0.15 dB per MZI and 0.025 dB/cm propagation, with 7 dB/facet edge coupling (reduced to 3.4 dB via grating couplers). A 4-mode, mesh dissipated 1.5 mW across 24 phase shifters with 600 MHz switching [168, 167]. Industry BTO-on-SiN reports 0.1 dB per phase element, 0.127 dB/facet coupling, 0.016 dB/cm propagation, and 6.9 GHz switching, though achievable mode scaling is unclear [169]. Another BTO platform reports a 4-mode mesh with 560 nW static per element at 12.5 MHz; scaled to a 4-layer Clements topology, 14 μ W static is expected, but current 1.48 dB/element loss is impractical for quantum computing applications [184].

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